Moon Formation by Triple Phase Transition in the Differentiating Proto-Earth

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Abstract. The origin of the Moon remains one of the unresolved problems of planetology. The canonical giant-impact model faces growing geochemical difficulties: it does not naturally predict the near-perfect Earth–Moon isotopic identity, the crustal dichotomy, or the ≈ 350 Myr delay of the terrestrial dynamo. This work proposes an alternative grounded in a necessary thermodynamic observation: every Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle, accretion energy exceeding the total melting energy by a factor of ≈ 155 (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The proto-Earth is therefore a rapidly rotating magmatic body ($T_{\mathrm{rot}} \approx 3.5$ h — a value constrained directly by the terrestrial-planet-accretion literature, without recourse to a contraction argument from an assumed initial period) with no stabilizing satellite.

In the Hadean context — a T-Tauri-type Sun whose coronal mass ejections are 10 to 100 times more intense than today (Feigelson & Montmerle, 1999; Airapetian et al., 2016), a dense silicate atmosphere at 2,000–4,000 K acting as a thermal blanket (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), continuous planetesimal bombardment (Agnor et al., 1999; Kokubo & Genda, 2010), and active short-lived radioactivity (Urey, 1955; Huss et al., 2009) — the proto-Earth's axis undergoes a large-amplitude free nutation in the range $[40^\circ, 70^\circ]$ in its own co-rotating frame (Laskar et al., 1993a; Laskar, 1993b; Touma & Wisdom, 1993; Li & Batygin, 2014). This is *not* astronomical obliquity relative to a nonexistent Hadean ecliptic; it is a purely internal geometric oscillation of the fluid body.

In this context, a single engine — the progressive segregation of Fe-Ni toward the forming core — drives three coupled transitions. The first is a rheological transition: the silicate magma acquires a Bingham-Herschel yield stress (Bingham, 1922; Herschel & Bulkley, 1926) and organizes a Coherent Magmatic Torus (CMT) within the intertropical band $|\phi| < 30^\circ$. The second is a mechanical transition: the CMT undergoes two to three episodes of cohesive hypersonic ejection beyond the Roche radius, governed by a double potential well and the stochastic Kramers crossing rate (Kramers, 1940). Each ejection alters the proto-Earth's inertia tensor and shifts the instantaneous rotation axis toward a new obliquity value within $[40^\circ, 70^\circ]$; since the intertropical band is defined perpendicular to this axis, this realignment shifts the active band itself, so that the next torus re-forms at a different location and on a spheroid whose own shape has changed — each successive CMT is therefore geometrically distinct from the previous one, underlying the observed near-side/far-side crustal dichotomy. The third is a magnetic transition: the terrestrial dynamo switches on ≈ 350 Myr after ejections cease, consistent with the paleomagnetic data from the Jack Hills zircons (Tarduno et al., 2025).

The central equation gives an ejected mass of 2.2–4.0 $\times 10^{22}$ kg per episode; with $N = 2$–3 episodes and a capture efficiency of 0.70, the lunar mass is reconstituted. The mechanism requires no external impactor and satisfies the constraints of isotopic identity, iron depletion, and angular momentum as direct mechanical consequences. The central prediction is one or more seismic interfaces between 200 and 530 km depth, with an impedance contrast $|R| \in [0.01, 0.04]$. This prediction is testable by **Chang'e 7** (South Pole, August 2026), **FSS** (Farside Seismic Suite), **LEMS** (Lunar Environment Monitoring Station), and **Artemis III** (2028–2029). Preliminary observational support is provided by Chang'e-6 samples (Yue et al., 2026; Xu et al., 2025): norites dated at 4247 ± 5 Myr (consistent with Episode 1) and anomalously high Fe/Mn ratios in deep olivines, both consistent with the predicted Fe/Si gradient (P3) and the South Pole Fe/Si anomaly (P6). The CMT's confinement to the intertropical band admits an effective description via a nonlinear Schrödinger equation, whose

1 Introduction: Why a New Theory of Lunar Formation?

1.1 The Giant-Impact Problem

The canonical giant-impact model (Cameron & Ward, 1976; Hartmann & Davis, 1975) dominated planetology for nearly fifty years. In this scenario, a Mars-sized body (Theia) struck the proto-Earth, and the resulting debris disk accreted to form the Moon. This model explains several first-order features: the angular momentum of the Earth–Moon system, the relatively small size of the lunar core, and the absence of a massive atmosphere.

However, over the past decade, growing geochemical and chronological difficulties have eroded confidence in the giant-impact scenario as a complete explanation:

1. **Earth–Moon isotopic identity**: the isotopic compositions of oxygen, titanium, tungsten, and chromium are nearly identical between Earth and the Moon to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The giant-impact model requires either that Theia had an isotopic composition identical to Earth's (an improbable coincidence at $< 1\%$), or complete post-impact isotopic mixing (Canup, 2012), which is not naturally predicted by hydrodynamic simulations.
2. **Iron depletion**: the Moon is significantly depleted in siderophile elements relative to the terrestrial mantle (O'Neill, 1991; Jones & Drake, 1986). The giant impact explains this depletion by assuming that Theia was pre-differentiated (with a metallic core that accreted into Earth), but this assumption adds a parameter.
3. **Crustal dichotomy**: the Moon's far side is thicker (≈ 54 km) and older than the near side (≈ 34 km) (Wieczorek et al., 2013). The giant impact does not predict this asymmetry.
4. **Delay of the terrestrial dynamo**: the terrestrial magnetic field is detected in Jack Hills zircons at 4.2 Ga (Tarduno et al., 2025), i.e. ≈ 350 Myr after the estimated formation of the Moon (4.55 Ga). The giant impact offers no explanation for this delay.

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in Moon formation (Sossi et al., 2025). This conclusion motivates the exploration of alternatives grounded in the internal dynamics of the proto-Earth.

1.2 The Triple Phase Transition: An Endogenous Alternative

This work proposes that the Moon formed not through an external impact, but through an endogenous process within the differentiating proto-Earth itself. The central thesis is as follows.

Hypothesis H0 (central thesis). The Moon formed through a series of *three coupled transitions* — rheological, mechanical, and magnetic — triggered by the single engine of Fe-Ni segregation toward the forming core. The result is a Moon built layer by layer in two to three episodes of cohesive magmatic ejection, each layer archiving the state of Hadean differentiation at the moment of its accretion.

The mechanism rests on a necessary thermodynamic observation: every Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). Accretion energy exceeds the total melting energy by a factor of ≈ 155. The proto-Earth is therefore a rapidly rotating magmatic body ($T_{\text{rot}} \approx 3.5$ h) with no stabilizing satellite.

In this rapidly rotating fluid body, the rotation axis undergoes a large-amplitude free nutation in the range $[40°, 70°]$ in the co-rotating frame (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014) — not astronomical obliquity, but an internal oscillation of the fluid body. This oblique tilt is the geometric condition that allows the radial Coriolis force ($f_{\text{rad}} = 2\Omega U \sin\varepsilon$) to destabilize the flow rather than confine it.

1.3 Organization of the Manuscript

The manuscript is organized as follows. Section 2 presents the Hadean physical context (rotation, obliquity, silicate atmosphere). Section 5 presents the intertropical band as the ejection site. Section 6 responds to the Taylor-Proudman objection. Section 7 inventories the forces. Section 8 explicitly ranks all hypotheses. Section 9 presents the three coupled transitions (rheological, mechanical, magnetic) with the complete resolution of the double well. Section 10 presents the mass budget. Section 11 presents angular momentum transport. Section 12 gives the 3–4 week chronology. Section 13 presents internal stratification. Section 14 discusses observational constraints and existing evidence. Section 15 presents falsifiable predictions. Section 16 presents the validation windows. Section 17 discusses the main objections. Section 18 presents the limitations. Section 19 concludes.

2 The Hadean Inferno: Physical Context

Before delving into the physics of the Triple Phase Transition, it is essential to understand the framework in which it operates. This framework is not a quiet void: it is one of the most energetically violent periods in the history of the Solar System.

2.1 The Dynamic Context of Terminal Accretion

The Hadean period (4.568–4.4 Ga) is characterized by a set of intense physical processes that define the initial conditions and the continuous forcings of the system:

- **Continuous planetesimal accretion**: the protoplanetary disk remains populated with planetesimals and embryos, providing constant bombardment (Agnor et al., 1999; Kokubo & Genda, 2010).
- **Rapid rotation**: the proto-Earth rotates with a period $\approx$ 3.5 h, a regime in which Coriolis and centrifugal effects dominate the internal dynamics.
- **Absence of a tidal stabilizer**: without a Moon, obliquity is free to evolve over a wide chaotic range.
- **A young and active Sun**: the T-Tauri-type Sun irradiates intensely and produces coronal mass ejections (CMEs) that perturb the system (Feigelson & Montmerle, 1999; Airapetian et al., 2016).
- **Short-lived radioactivity**: the isotopes ^{26}Al and ^{60}Fe are still active, injecting heat into the mantle (Urey, 1955; Huss et al., 2009).
- **Dense silicate atmosphere**: an envelope of rock vapor at $T \sim 2,000$–$4,000$ K and $P \sim 10^5$–10^7 Pa constitutes a thermal blanket and a lateral confinement agent (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013).

The combination of these processes — not any one of them in isolation — defines the physical environment in which the TPT operates. The T-Tauri Sun, although important, is just one ingredient among others in this dynamic context.

2.2 The Silicate Atmosphere: Thermal Blanket and Lateral Confinement

At $t = 0$, the proto-Earth is enveloped in a dense silicate atmosphere — rock vapor, gaseous SiO, Mg and Fe — at temperatures of 2,000 to 4,000 K and pressures of 10^5–10^7 Pa. This cocoon plays two critical roles in the TPT.

First role: thermal insulation. The silicate atmosphere has a strong infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). Without this atmosphere, the proto-Earth would have cooled within $\sim 10^4$ years. With it, the melting window extends to 30–50 Myr — exactly the timescale of core formation (Kleine et al., 2002; Yin et al., 2002). This is not an ad hoc hypothesis: it is the physical consequence of a rock-vapor atmosphere at $2,000$–$4,000$ K. The same mechanism operates in magma-ocean models (Elkins-Tanton, 2008; Lebrun et al., 2013).

Second role: lateral confinement. The confining pressure ($P \approx 10^5$–10^7 Pa) opposes the lateral expansion of the CMT flow during ejection. It guarantees the cohesion of the ejected sheet. The high speed of sound in this torrid atmosphere ($c_{\text{sound}} \approx 1,500$ m s^{-1} at $T_{\text{atm}} \approx 3,000$ K) reduces the effective Mach number of the flow, improving cohesion relative to ejection into vacuum.

Without this atmosphere, the melting window would close within $\sim 10^4$ years. With it, the window extends to 30–50 Myr (Elkins-Tanton, 2008; Lebrun et al., 2013;

Hamano et al., 2013), corresponding to the timescale of core formation (Kleine et al., 2002; Yin et al., 2002).

2.3 Continuous Planetesimal Bombardment

The protoplanetary disk remains populated with planetesimals and embryos during the TPT's active window (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2001). Each impact: (1) reinjects thermal energy into the mantle, sustaining melting; (2) generates a spherical pressure wave propagating through the entire melt, perturbing $U(r)$ and stochastically contributing to Kramers barrier crossings; (3) delivers an impulsive torque on the spin vector, resetting the axial oscillation.

2.4 Short-Lived Radioactivity: Source of Internal Heat

The isotopes ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) are still active during the first million years (Urey, 1955; Huss et al., 2009), injecting $\approx 3 \times 10^{29}$ J of additional heat into the mantle and contributing to sustained melting throughout the TPT's window.

3 Hadean Obliquity $\varepsilon \in [40^\circ, 70^\circ]$: A Fundamental Geometric Clarification

The range $\varepsilon \in [40^\circ, 70^\circ]$ is not a free parameter. It is constrained by five independent and convergent physical arguments. Before enumerating them, a fundamental geometric clarification is necessary.

Geometric clarification. In this work, ε denotes the instantaneous angle between the rotation vector $\mathbf{\Omega}(t)$ and the principal axis of inertia of the Maclaurin spheroid, **defined in the body's co-rotating frame**. This is *not* astronomical obliquity relative to the ecliptic plane: that concept has no physical meaning for the Hadean proto-Earth, which has neither a fixed reference ecliptic nor a Moon. It is a purely internal geometric angle describing the *oscillation* — the free nutation — of the rotation axis relative to the body's own shape.

Definition of the intertropical band. The Hadean intertropical band $|\phi| < 30^\circ$ is defined perpendicular to the *instantaneous* rotation axis $\mathbf{\Omega}(t)$, not relative to a fixed geographic frame. The plane $\phi = 0$ is therefore the *instantaneous dynamical equator* of the body, which oscillates together with the free nutation of $\mathbf{\Omega}(t)$ — it is not a fixed geographic equator. The coordinate ϕ is the oblique latitude measured from this instantaneous equator. This is the plane where the radial Coriolis force $f_{\mathrm{rad}} = 2\Omega U \sin\varepsilon$ is maximal and where the effective gravity $g_{\mathrm{eff}}(\phi)$ is minimal (Section 4.3). The band is therefore a dynamical structure tied to the instantaneous equator, not a fixed latitude. This identification of ϕ with the instantaneous dynamical equator, rather than a fixed geographic one, is used directly in Section 7.3 and again in the derivation of Appendix C.

Argument 1 — Absence of a tidal stabilizer (fundamental). The proto-Earth has no Moon. Without a massive satellite, there is no lunar tidal torque to damp the precession of the spin axis and keep it near a stable equilibrium. This argument is logically prior to all the others: it is precisely the Moon whose formation we seek to explain, so invoking it as a stabilizer would be circular. The absence of a tidal stabilizer is therefore a necessary, not contingent, condition of the Hadean proto-Earth.

Argument 2 — Laskar et al. (1993): a chaotic zone without a satellite. Laskar et al. (1993a) showed that without its Moon, Earth's obliquity would evolve chaotically over a zone extending from nearly 0° to $\approx 85^\circ$. Although this result strictly applies to a present-day Earth rotating at ≈ 24 h, it establishes the general principle: *the Moon is the stabilizer, and without it, large obliquity excursions are the norm*. For the Hadean proto-Earth, this principle applies *a fortiori*.

Argument 3 — Rapid rotation at $T_{\mathrm{rot}} = 3.5$ h. The precession constant satisfies $\alpha_{\mathrm{prec}} \propto \omega$. At $T_{\mathrm{rot}} = 3.5$ h, the precession frequency is $\approx 7\times$ higher than at 24 h, pushing the secular spin-orbit resonances identified by Laskar et al. (1993a) out of reach. Li & Batygin (2014), working with a rapidly rotating Moonless Earth, found that obliquity remains chaotic but *diffuses slowly* over a wide range. Rapid rotation does not stabilize the axis — it simply changes the nature of the chaos, from resonant to diffusive.

Argument 4 — Continuous planetesimal bombardment. Each major impact during terminal accretion (Agnor et al., 1999) abruptly resets the spin vector. The relaxation time toward axial alignment for an isolated fluid body,

$$\tau_{\mathrm{relax}} \sim \frac{\eta\, R_{\mathrm{eq,proto}}^3}{G\, M_\oplus^2\, f} \approx 10^6\,\mathrm{yr} \quad (\eta \sim 10^{1\text{–}3}\,\mathrm{Pa\,s},\ f \approx 0.107), \tag{1}$$

far exceeds the interval between impacts (10^3–10^5 yr). Misalignment is continuously maintained: $\tau_{\mathrm{pert}} \ll \tau_{\mathrm{relax}}$.

Argument 5 — T-Tauri CME torques. As described in Section 7.4, the impulsive torques from CMEs act on timescales of days to weeks, much shorter than τ_{relax}. They constitute a continuous stochastic source of obliquity perturbation (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

The range $\varepsilon \in [40^\circ, 70^\circ]$ is not an ad hoc hypothesis. It is the predictable consequence of the dynamics of a rapidly rotating fluid body, lacking a tidal stabilizer, subject to continuous stochastic perturbations. The simulations of Laskar et al. (1993a) for a Moonless Earth, and of Li & Batygin (2014) for a rapidly rotating Moonless Earth, converge on this range — the same one adopted by Laskar et al. (1993a) and Touma

& Wisdom (1993). In this oblique geometry, the Coriolis force decomposes into radial and horizontal components (Section 7.3). The radial component $f_{\mathrm{rad}} = 2\Omega \sin \varepsilon \cdot U$ is maximal for $\varepsilon \in [40^\circ, 70^\circ]$.

Section Summary

Five convergent arguments support $\varepsilon \in [40^\circ, 70^\circ]$. The absence of a tidal stabilizer is logically necessary. Arguments 2 through 5 provide independent physical justification for a wide and sustained misalignment. The range $[40^\circ, 70^\circ]$ is adopted as the physically motivated Hadean range; the mechanism operates for any $\varepsilon > 45^\circ$ (Section 7.3). The intertropical band $|\phi| < 30^\circ$ is tied throughout this work to the instantaneous dynamical equator $\mathbf{\Omega}(t)$, not to a fixed geographic frame.

3.1 Relaxation Time and Rheological Reinforcement

The relaxation time toward axial alignment for an isolated fluid body is given by Eq. (1). Bingham-Herschel rheology further slows this relaxation. The rheological response time to a stress perturbation is:

$$\tau_{\mathrm{BH}} \sim \frac{\tau_y}{\mu \dot{\gamma}_{\mathrm{typ}}} \approx 10^2\text{–}10^4\ \mathrm{s}, \tag{2}$$

where μ is the dynamic viscosity and $\dot{\gamma}_{\mathrm{typ}}$ the typical shear rate. This time is much shorter than τ_{relax}, which means that mass redistribution is locally frozen by the yield stress, further delaying axial alignment.

4 Initial State of the Proto-Earth

4.1 Near-Total Melting: A Necessary Thermodynamic Consequence

The formation of an Earth-mass planet releases an enormous amount of gravitational energy. In the homogeneous-sphere approximation (Safronov, 1969), the total accretion energy is:

$$E_{\mathrm{grav}} = \frac{3\, G\, M_\oplus^2}{5\, R_{\mathrm{eq,proto}}} \approx 2.49 \times 10^{32}\ \mathrm{J}, \tag{3}$$

where G is the gravitational constant, $M_\oplus$ the mass of Earth, and $R_{\mathrm{eq,proto}} \approx 7.4$ Mm the equatorial radius of the undifferentiated proto-Earth.

Equation (3) is a thermodynamic bound, not a model of instantaneous formation. It computes the *total* gravitational energy released during accretion, independent of its duration. This is the standard approach in the literature (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The manuscript explicitly cites Kleine et al. (2002); Yin et al. (2002) as evidence that accretion lasted ~ 30 Myr.

The energy required to melt the entire silicate mantle (Stixrude & Lithgow-Bertelloni, 2014):

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24}\,\text{kg} \times 4 \times 10^{5}\,\text{J}\,\text{kg}^{-1} \approx 1.6 \times 10^{30}\,\text{J}, \tag{4}$$

where M_{mantle} is the mass of the silicate mantle and L_{fus} the specific latent heat of fusion.

The energy ratio is a consensus result:

$$E_{\text{grav}}/E_{\text{fus}} \approx 155. \tag{5}$$

This is established by Solomatov (2000), Elkins-Tanton (2012), and Rubie et al. (2015). Accretion energy exceeds the total melting energy by two orders of magnitude.

Why Melting Was Sustained Despite Radiative Cooling

Accretion lasted ~ 30 Myr (Kleine et al., 2002; Yin et al., 2002). During this period, radiative cooling occurred. Four independent mechanisms explain why melting was sustained despite this cooling.

Mechanism 1 — The silicate atmosphere as a thermal blanket. The dense silicate atmosphere ($P \approx 10^5$–10^7 Pa, $T \approx 2,000$–$4,000$ K) has a strong infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). The effective cooling time goes from $\sim 10^4$ years to 30–50 Myr.

Mechanism 2 — Additional heat sources. Four heat sources extend melting beyond the initial accretion energy:

- **Fe-Ni segregation heat:** $\approx 3 \times 10^{30}$ J (Flasar & Birch, 1973; Rubie et al., 2003).
- **Short-lived radioactivity:** ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) inject $\approx 3 \times 10^{29}$ J (Urey, 1955; Huss et al., 2009).
- **Continuous planetesimal bombardment:** reinjection of thermal energy (Agnor et al., 1999; Kokubo & Genda, 2010).
- **T-Tauri irradiation:** maintains surface temperatures above 2,000 K (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

Mechanism 3 — The TPT does not require melting of the entire mantle. The Triple Phase Transition requires melting only within the Hadean intertropical band $|\phi| < 30^\circ$ (upper and middle mantle). The factor of 155 thermodynamically guarantees this partial melting, even if the deep mantle were not entirely molten (Nomura et al., 2011).

Mechanism 4 — The TPT's active window is 30–50 Myr. The TPT occurs *during* the 30–50 Myr window over which melting is sustained by the mechanisms above. Core formation over ~ 30 Myr (Kleine et al., 2002; Yin et al., 2002) is subsequent to, or simultaneous with, this — it is not a contradiction.

Near-total melting is the mandatory starting point. The objection based on the ~ 30 Myr accretion timescale does not contradict the mechanism: the TPT occurs *during* the melting window sustained by the silicate atmosphere, the additional heat sources, and continuous bombardment.

Hypothesis H1 — Total volumetric melting. At the onset of Fe-Ni segregation, the entire silicate mantle lies above the liquidus, with no stable internal solid interface during the active window 4.568–4.4 Ga. This is a self-gravitating fluid volume, not a magma ocean resting on a substrate: the dynamics, the rheology, and the response to perturbations are physically distinct.

Caveat: Nomura et al. (2011) suggest that the deep lower mantle might not have been entirely molten, owing to the rise of the solidus with pressure. The TPT requires melting only within the Hadean intertropical band $|\phi| < 30°$ (upper and middle mantle), which the factor of 155 thermodynamically guarantees. The adiabatic temperature profile (Appendix A) confirms melting throughout the entire mantle.

4.2 Initial Rotation Period: $T_{\mathrm{rot}} \approx 3.5$ h

The proto-Earth's initial rotation period is not a free parameter: it is constrained directly by the terrestrial-planet-formation literature to $T_{\mathrm{rot}} \approx 3$–5 h.

1. T-Tauri magnetic torque. The young, rapidly rotating Sun (≈ 3 days, Bouvier et al. 2014) magnetically couples to the inner protoplanetary disk (Königl, 1991; Shu et al., 1994), transferring angular momentum to material in the inner orbit and accelerating the proto-Earth.

2. N-body simulations of terrestrial planet formation. N-body simulations show that the final spin periods of Earth-mass planets are in the range of 3–5 h (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2013). Agnor et al. (1999) find that protoplanetary embryos are spinning rapidly throughout the final stages of accretion, upon suffering their first collision, suggesting that the proto-Earth may have been rotating rapidly prior to any subsequent large impact. Kokubo & Genda (2010), using N-body simulations of protoplanet formation under realistic accretion conditions, find final rotation periods of 3–5 h for Earth-mass bodies, consistent with Chambers (2013). The value $T_{\mathrm{rot}} \approx 3.5$ h adopted here lies within this independently established range, and is of the same order of magnitude as the rapid pre-impact rotations considered in the giant-impact models of Ćuk & Stewart (2012) (2.3–2.7 h) and Canup (2012) (a pre-impact rotation of 3 h in one of several simulated cases) — though neither of these specific values is identical to 3.5 h, nor does the TPT rely on the giant-impact hypothesis itself.

Why a rapid rotation requires a spin-up mechanism. Dones & Tremaine (1993) show analytically that ordered accretion from a uniform or slowly varying disk of small bodies cannot produce a rotation as rapid as that of Earth or Mars, for any value of the disk's velocity dispersion: the prograde and retrograde contributions of individual planetesimals nearly cancel by symmetry. The rapid spin rates of the terrestrial planets are therefore most naturally explained by one or a few collisions between large bodies late in accretion — a requirement that the TPT satisfies through the continuous planetesimal bombardment of Section 2, independently of whether a single dominant impactor (as in the giant-impact hypothesis) is invoked. Consistently with this, Kokubo & Ida (2007), using N-body simulations of the embryo-merging stage of accretion, find that planetary spin angular velocities follow a Maxwellian distribution with substantial dispersion: there is no single privileged rotation period predicted by accretion physics alone, only a broad statistical range within which $T_{\mathrm{rot}} \approx 3.5$ h is a plausible value.

3. Skater effect: Fe-Ni segregation. The migration of Fe-Ni toward the forming core reduces the moment of inertia as segregation proceeds. With $I_f / I_0 \approx 0.82$ (Rubie et al., 2015), this contributes a further $\approx 18\%$ acceleration of rotation over the course of the TPT window, on top of whatever rotation rate the proto-Earth already had from accretion. This effect is qualitative and contributory: it is not used here to derive $T_{\mathrm{rot}} \approx 3.5$ h from an assumed pre-segregation value, since no such value is independently constrained.

4. Distinction from the synestia regime. Lock & Stewart (2017) define the corotation limit (CoRoL) as the rotation rate at which the equatorial velocity of a body equals the local Keplerian orbital velocity; bodies exceeding this limit form a distended, partially-vaporized structure they term a synestia, rather than a body in simple rigid rotation. For the proto-Earth's parameters adopted here ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm), the corotation limit corresponds to a period of ≈ 1.8 h. At $T_{\mathrm{rot}} \approx 3.5$ h, Ω is only $\approx 50\%$ of this limit: the proto-Earth described by the TPT is therefore well within the regime of a body in coherent rigid rotation, not a synestia.

On the Initial Rotation Period

The value $T_{\rm rot} \approx 3.5$ h is constrained directly by the terrestrial-planet-formation literature (Agnor et al. 1999; Kokubo & Genda 2010; Chambers 2013; 3–5 h), is of the same order of magnitude as the rapid rotations considered in giant-impact models (Ćuk & Stewart, 2012; Canup, 2012) without depending on the giant-impact hypothesis, and is consistent with the theoretical requirement that rapid terrestrial-planet spin demands a late-accretion spin-up mechanism (Dones & Tremaine, 1993), within the broad statistical range found for planetary spin in N-body simulations (Kokubo & Ida, 2007). It is not derived from an assumed initial period via a contraction argument. The Coriolis compensation parameter b', a simple ratio between the critical ejection velocity $U_{\rm crit}$ and the geostrophic equilibrium velocity of the CMT, provides an independent cross-check: Fe-Ni segregation increases b', ejections decrease it, and the equilibrium point $b' = 1$ is a stable attractor that naturally selects $T_{\rm rot} \approx 3.5$ h and $\varepsilon \approx 55°$. The complete derivation is given in Appendix B.

4.3 Geometry of the Proto-Earth: The Maclaurin Spheroid

A rapidly rotating fluid mass takes the shape of a Maclaurin spheroid (Chandrasekhar, 1969). The equatorial radius is $R_{\rm eq,proto} \approx 7.4$ Mm, combining reduced density (+4%), thermal expansion (+7.5%), and equatorial bulging (+7.5%). The dynamical flattening:

$$J_2 \approx \frac{5}{4} \cdot \frac{\Omega^2 R_\oplus}{g_0} \approx 0.101, \tag{6}$$

where g_0 is the mean surface gravity, gives an equatorial bulge $\delta R \approx 642$ km. Effective gravity varies with latitude ϕ (measured from the instantaneous dynamical equator, as defined in Section 3) as:

$$g_{\rm eff}(\phi) = g_{\rm eff}^{(0)} \left(1 + \tfrac{3}{2} J_2 \sin^2\phi\right) - \Omega^2\, r(\phi) \cos^2\phi, \tag{7}$$

where $r(\phi)$ is the local radius. Effective gravity is minimal within the band $|\phi| < 30°$: this is the CMT's geometric attractor.

The free (Euler) nutation period is:

$$\omega_E = \Omega f \approx 3.74 \times 10^{-5}\,\mathrm{rad\,s^{-1}}, \qquad T_{\rm nut} \approx 46.7\,\mathrm{h}. \tag{8}$$

This frequency plays a central role in the parametric elliptical instability (Section 7.6).

Stability of the Maclaurin spheroid at this rotation rate. A rapidly rotating self-gravitating fluid body need not adopt the axisymmetric Maclaurin form: beyond a critical rotation rate, the Maclaurin sequence bifurcates into the triaxial Jacobi sequence. The bifurcation criterion compares two characteristic timescales: the self-gravity timescale, $t_g \sim 1/\sqrt{G\bar\rho}$, and the rotation period, $t_{\rm rot} = 2\pi/\Omega$; their ratio defines the dimensionless number $\Omega^2/(\pi G\bar{\rho})$. This criterion is derived analytically by Chandrasekhar (1969) (chapter 4, full demonstration; see also chapters 3 to 5 for Maclaurin spheroids and chapters 6 and 7 for Jacobi ellipsoids and the bifurcation itself) as the point where the family of Jacobi ellipsoids branches off the Maclaurin sequence — not as an empirical approximation — and is presented from a historical perspective by Jeans (1929), taken up in the stellar context by Tassoul (1978), the theory being formally identical for any self-gravitating fluid; Lamb (1932) gives the classical hydrodynamic treatment, and Durand (1964) the ellipsoidal integrals on which Chandrasekhar's demonstration rests. This bifurcation occurs at $\Omega^2/(\pi G\bar{\rho}) \approx 0.374$, with Maclaurin spheroids becoming genuinely unstable only beyond $\Omega^2/(\pi G\bar{\rho}) \approx 0.440$.

The ρ of this criterion is the uniform density of the homogeneous fluid in the theoretical model. Its application to a real planetary body substitutes for this uniform density the mean density $\bar{\rho} = M/(\frac{4}{3}\pi R^3)$, an approximation explicitly discussed by Chandrasekhar (1969) (chapters 1 and 5) and Tassoul (1978) (chapter 2). For the Hadean proto-Earth, $\bar{\rho}$ is not directly measurable: it depends on the degree of differentiation already reached, which is itself not precisely known. Available estimates place the mean density of a homogeneous Earth of the same mass, prior to complete core segregation, around 4,100–4,500 $\mathrm{kg\,m^{-3}}$, against 5,510 $\mathrm{kg\,m^{-3}}$ for the fully differentiated present-day Earth (Rubie et al., 2015; Trønnes et al., 2019; Rubie et al., 2016; Hirose et al., 2021). These values are themselves derived from mineralogical models (chondritic or pyrolitic composition, equations of state) rather than direct measurement, and so constitute a modeling range rather than a physical constant.

For the proto-Earth's parameters adopted here ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm, mean density $\bar{\rho} \approx 4{,}300\ \mathrm{kg\,m^{-3}}$, the median value of this range, for a still weakly differentiated body), $T_{\mathrm{rot}} \approx 3.5$ h gives $\Omega^2/(\pi G\bar{\rho}) \approx 0.276$ — below the Jacobi bifurcation point (0.374), with a margin of roughly 26%. At this density, the bifurcation threshold itself corresponds to $\Omega_c = \sqrt{0.374\,\pi G\bar{\rho}} \approx 5.8 \times 10^{-4}\ \mathrm{rad\,s^{-1}}$, i.e. $T_{\mathrm{rot}} \approx 3.0$ h, and the strict dynamical-instability threshold to $T_{\mathrm{rot}} \approx 2.8$ h. The axisymmetric Maclaurin geometry adopted throughout this work is therefore consistent with the rotation rate of the TPT, and does not require recourse to a triaxial or otherwise more complex equilibrium figure; the available margin nonetheless remains sensitive to the exact value of $\bar{\rho}$ adopted, which is not known to better precision than the modeling range cited above.

5 The Intertropical Band $|\phi| < 30^\circ$: The Ejection Site

A fundamental and often misunderstood point: the CMT is not an equatorial structure. It is a coherent zonal band extending over $\pm 30^\circ$ of oblique latitude — roughly one third of the proto-Earth's surface. This extent results from three independent and convergent mechanisms.

As established in Section 3, the band $|\phi| < 30°$ is defined perpendicular to the instantaneous rotation axis $\mathbf{\Omega}(t)$, not relative to a fixed geographic frame. This is the plane where the radial Coriolis force is maximal and the effective gravity is minimal.

Mechanism 1 — Maclaurin geometry. Effective gravity is minimal within the band $|\phi| < 30°$ (Eq. 7). Molten material accumulates there through centrifugal effects, independently of any other argument. This is a consequence of the ellipsoidal shape imposed by rapid rotation.

Mechanism 2 — The geostrophic inverse cascade. The CMT's Rossby number is deeply geostrophic:

$$\mathrm{Ro} = \frac{U_{\mathrm{turb}}}{2\,\Omega\,R_{\mathrm{eq,proto}}} \approx 1.35 \times 10^{-4} \ll 1, \tag{9}$$

In this regime, turbulent energy cascades toward large scales (inverse cascade) (Rhines, 1975). The Rhines scale:

$$L_\beta = \sqrt{\frac{U_{\mathrm{turb}}}{\beta}}, \qquad \beta = \frac{2\Omega\cos\varepsilon}{R_{\mathrm{eq,proto}}}, \tag{10}$$

gives $L_\beta \approx 0.52\,R_\oplus$ — corresponding to the harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible on a sphere. The CMT is the statistical attractor of this cascade (Rhines, 1975; Sukoriansky et al., 2007).

Mechanism 3 — The radial Coriolis force. $f_{\mathrm{rad}} = 2\Omega U \sin\varepsilon$ concentrates the flow within the oblique intertropical band and keeps it coherent there.

Section Summary

These three mechanisms — Maclaurin geometry, the geostrophic inverse cascade, and the radial Coriolis force — are independent and converge on the same band $|\phi| < 30°$. Their simultaneous convergence is an argument in favor of the theory's structural robustness.

6 The CMT as a Purely Toroidal Flow: Response to the Taylor-Proudman Objection

Any geophysical fluid dynamicist confronted with the CMT will naturally raise the following objection: in a rapidly rotating fluid (Ro $\ll 1$), the Taylor-Proudman theorem imposes $(\mathbf{\Omega} \cdot \nabla)\boldsymbol{u} = 0$, generating column structures parallel to $\mathbf{\Omega}$, rather than a single oblique band $|\phi| < 30°$. The objection is legitimate and deserves a precise response.

6.1 The Poloidal/Toroidal Decomposition

Any velocity field within a spherical shell decomposes into a poloidal part (radial and meridional components) and a toroidal part (purely azimuthal component):

$$\boldsymbol{u} = \boldsymbol{u}_{\text{pol}} + \boldsymbol{u}_{\text{tor}}, \qquad \boldsymbol{u}_{\text{tor}} = \nabla \times (\psi \hat{\boldsymbol{r}}).$$

The Taylor-Proudman theorem, in its classical form, constrains the poloidal field (which carries vorticity along $\boldsymbol{\Omega}$) and produces Taylor columns. It does not constrain the purely toroidal field.

For a purely azimuthal flow independent of longitude ($u_r = u_\theta = 0$ on average, $u_\phi = u_\phi(r, \phi)$), the Taylor-Proudman constraint is automatically satisfied:

$$(\boldsymbol{\Omega} \cdot \nabla) u_{\text{tor}} = 0 \tag{11}$$

for any toroidal field, with no additional condition. The equilibrium is cyclostrophic, not geostrophic: the centrifugal pressure gradient balances the Coriolis force within the azimuthal plane.

6.2 Why Curved Columns Do Not Extend Beyond the Active Band

In an obliquely rotating spherical shell ($\varepsilon \neq 0$), Taylor columns are curved along the projected field lines of $\boldsymbol{\Omega}$. They tend to channel the flow away from its source — unless a cutoff mechanism stops them.

This is precisely the role of the yield stress τ_y. In a Bingham-Herschel fluid, flow propagates only in regions where the stress exceeds τ_y. As soon as the columns enter a region where $\tau < \tau_y$, they are truncated: the magma solidifies there (Frigaard & Nouar, 2017; Balmforth & Craster, 2001). The band $|\phi| < 30°$ is precisely the region where effective gravity is minimal (Maclaurin) and where the Coriolis stress is maximal ($\varepsilon \approx 55°$): it is there, and only there, that $\tau > \tau_y$. The curved columns arising from the active band are thus naturally truncated at its boundaries.

Section Summary

The CMT is a purely toroidal flow. The Taylor-Proudman objection does not apply. What remains to be demonstrated numerically — identified here as Limitation L1 (Section 18) — is whether a single torus forms within the band $|\phi| < 30°$ rather than several alternating bands. This is a question for simulation, not a theoretical inconsistency.

7 Complete Inventory of Forces

All forces, projections, and equations are expressed in the body's instantaneous Hadean co-rotating frame, with ε defined as in Section 3.

7.1 Complete Inventory of Forces: Summary Table

The CMT is a toroidal flow in a rapidly rotating fluid body, subject to a set of forces. Table 1 summarizes all the forces, their physical role, and their mathematical expression.

Table 1: Complete inventory of forces acting on the CMT.

Force	**Role**	**Expression**
Effective gravity	Radial confinement	$g_{\text{eff}}(\phi) = g_{\text{eff}}^{(0)}(1 + \frac{3}{2}J_2 \sin^2\phi) - \Omega^2 r(\phi)\cos^2\phi$
Centrifugal force	Lateral ejection, flattening	$F_{\text{cent}} = \Omega^2 r(\phi)\cos^2\phi$
Radial Coriolis	Destabilization, outward ejection	$F_{\text{cor,rad}} = 2\Omega U \sin\varepsilon$
Horizontal Coriolis	Lateral confinement	$F_{\text{cor,hor}} = 2\Omega U \cos\varepsilon$
Taylor-Proudman	Vertical columns, toroidal flow	$(\boldsymbol{\Omega}\cdot\nabla)\boldsymbol{u}_{\text{pol}} = 0$; $\boldsymbol{u}_{\text{tor}}$ unconstrained
Planetesimal bombardment	Stochastic perturbation, thermal reinjection	$\delta\boldsymbol{\Omega} = \sum_i \delta\boldsymbol{\Omega}_i$, $\delta U = \sum_i \delta U_i$
T-Tauri (CME)	Stochastic perturbation, thermal maintenance	$\boldsymbol{F}_{\text{CME}}(t) = \sum_j \boldsymbol{F}_{\text{CME},j}\,\delta(t - t_j)$
Solar tide	Low-frequency forcing on obliquity	$F_{\text{tide,sol}} \sim 10^{-4} g_0 \cos(\omega_{\text{sol}} t)$
Proto-lunar tide	Mathieu resonance (episodes 2+)	$F_{\text{tide,moon}}(t) = A_m^{(k)}(t)\cos(n_{\text{Moon}} t)$
Euler acceleration	Axial nutation	$\boldsymbol{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}$
Atmospheric pressure	Lateral confinement	$P_{\text{atm}} \sim 10^5 - 10^7$ Pa
BH yield stress	Plasticity threshold	$\tau = \tau_y + k\dot{\gamma}^n$, $\tau > \tau_y$
Viscous stress	Dissipation	$\tau_{\text{visc}} = \eta\nabla^2\boldsymbol{u}$

7.2 Oscillation as a Direct Source of Force

Axial oscillation implies $\dot{\boldsymbol{\Omega}} \neq \boldsymbol{0}$. Poincaré (1910) showed that this generates an Euler acceleration on every fluid particle:

$$\boldsymbol{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}, \tag{12}$$

where $\boldsymbol{r}$ is the position vector. The radial projection:

$$\boldsymbol{a}_{\text{Euler}} \cdot \hat{r} = |\dot{\Omega}|\, r(\phi)\, \sin(\varepsilon + \phi), \tag{13}$$

is maximal within the band $|\phi| < 30^\circ$ where $r(\phi)$ is largest: the oscillation is therefore not mere decoration but a *direct energy source* for the CMT.

7.3 Radial Coriolis Force and Geometric Threshold

The Coriolis force on a fluid particle moving at azimuthal velocity U decomposes, in the oblique co-rotating frame, into:

$$f_{\text{rad}} = 2\,\Omega\, \sin\varepsilon \cdot U \quad \text{(radial, outward, destabilizing)}, \tag{14}$$

$$f_{\text{horiz}} = 2\,\Omega\, \cos\varepsilon \cdot U \quad \text{(horizontal, confining)}, \tag{15}$$

with the ratio:

$$\frac{f_{\text{rad}}}{f_{\text{horiz}}} = \tan\varepsilon. \tag{16}$$

Geometric threshold $\varepsilon = 45^\circ$ — necessary condition for radial ejection. For $\varepsilon > 45^\circ$: $\tan\varepsilon > 1$, the destabilizing radial component dominates the confining horizontal component. For $\varepsilon < 45^\circ$: the horizontal component dominates and the Taylor-Proudman constraint inhibits radial ejection. This threshold follows solely from Coriolis kinematics: no adjustable parameter is involved. At $\varepsilon = 55^\circ$: $f_{\text{rad}}/f_{\text{tot}} = \sin 55^\circ \approx 0.82$, i.e. 82% of the total Coriolis force directed radially outward.

7.4 The T-Tauri Sun: Catalyst and Perturber

The Sun in its T-Tauri phase is not a driving force on the same footing as gravity or Coriolis — it does not act continuously on every fluid parcel of the CMT. It nevertheless plays three essential roles within the TPT's window.

(1) Rheological modulator. T-Tauri UV/X-ray radiation keeps T_{surf} within the range $1,800$–$4,200$ K (Stefan-Boltzmann law, albedo $A_0 \approx 0.1$), which keeps the yield stress τ_y of the silicate magma within the range 10^2–10^4 Pa (Giordano et al., 2008) — the optimal window for CMT cohesion. Below $T^* \approx 1,800$ K, τ_y exceeds 10^6 Pa and cohesive ejection becomes physically inaccessible. The T-Tauri Sun thus acts as a *guardian of the rheological window* via surface radiative forcing transmitted to the mantle by conduction through the silicate atmosphere.

(2) Source of stochastic perturbations. CMEs (10^{27}–10^{29} J per event, several per day) and radiative pulsations provide $\sim 10^9$–10^{10} opportunities to cross the potential barrier (Section 9.2.2) over 10^6 yr. The number of opportunities is sufficient that the probability of no crossing whatsoever is negligible. CMEs act as *stochastic triggers* of the first ejection episode, once the dynamical conditions are met.

(3) Obliquity perturber. CMEs acting on the proto-atmosphere and the magnetosphere deliver impulsive torques on the spin vector, contributing to the maintenance of the axial oscillation $\varepsilon \in [40°, 70°]$ (Section 3). The T-Tauri phase also provides an independent stopping mechanism: once the Sun reaches the main sequence ($\approx$ 4.4 Ga), irradiation declines, τ_y increases, and the TPT window closes.

The T-Tauri Sun is therefore not the *engine* of the mechanism — gravity, Coriolis, the centrifugal force, Fe-Ni segregation, and the Taylor-Proudman columns are — but it is the *catalyst* that keeps the window open and provides the perturbations needed for stochastic crossings.

7.5 Tidal Forces

Solar tide. Relative amplitude $\approx 10^{-4} g_0$; contributes as a low-frequency resonant forcing on obliquity.

Growing proto-lunar tide — Mathieu resonance. From Episode 2 onward, the proto-satellite exerts a tidal force that modulates the height of the potential barrier:

$$\Delta E^{(k)}_{\mathrm{barr}}(t) = \Delta E^{(k,0)}_{\mathrm{barr}} - A^{(k)}_m(t)\cos(n_{\mathrm{Moon}}\, t), \tag{17}$$

where $A^{(k)}_m(t)$ is the growing tidal amplitude and n_{Moon} the proto-lunar mean motion. This gives the radial perturbation equation:

$$\ddot{\xi}_r = \left[\lambda + A^{(k)}_m(t)\cos(n_{\mathrm{Moon}}\, t)\right]\xi_r, \tag{18}$$

which is the Mathieu equation (McLachlan, 1947). Parametric resonance bands exist for $n_{\mathrm{Moon}} \approx 2\sqrt{|\lambda|}$ even when $\lambda < 0$: the growing proto-lunar tide is the dominant trigger for Episodes 2 through N.

Giant-planet tides. Mainly Jupiter: a low-frequency perturbation on obliquity, sustaining the oscillation over timescales of 10^5–10^6 yr.

7.6 Parametric Elliptical Instability as a CMT Amplifier

In a precessing fluid ellipsoid, the precessional flows can resonate with the fluid's inertial modes (Poincaré waves, frequencies $\omega_{\mathrm{in}} = 2\Omega m/n$). This mechanism — the parametric elliptical instability — was established theoretically by Malkus (1968), formalized by Kerswell (2002), and demonstrated experimentally by Lacaze et al. (2004).

For the Maclaurin spheroid geometry at $T_{\mathrm{rot}} = 3.5$ h and $f = 0.107$, the free nutation frequency is $\omega_E = \Omega f \approx 3.74 \times 10^{-5}$ rad s^{-1} (Eq. 8). This frequency is commensurable with the fluid's inertial modes by geometric construction — not by parameter tuning. The linear growth rate is:

$$\sigma_{\text{res}} = \frac{\Omega f}{2} \approx 1.87 \times 10^{-5}\,\text{rad}\,\text{s}^{-1}. \tag{19}$$

The CMT velocity grows exponentially:

$$U_{\text{CMT}}(t) = U_{\text{turb}}\, e^{\sigma_{\text{res}} t}, \qquad U_{\text{turb}} \approx 0.8\,\text{km}\,\text{s}^{-1}, \tag{20}$$

where U_{turb} is the characteristic velocity at the Rhines scale (Rhines, 1975). The time to reach $U_{\text{crit}} \approx 9.8$ km s^{-1} is:

$$t_{\text{acc}} = \frac{\ln(U_{\text{crit}}/U_{\text{turb}})}{\sigma_{\text{res}}} \approx 32\,\text{h}. \tag{21}$$

The CMT reaches the bifurcation velocity in $\approx$ 32 hours. This result is fully determined by $f = 0.107$ and $\Omega = 4.99 \times 10^{-4}$ rad s^{-1}, imposed by $T_{\text{rot}} = 3.5$ h and the physics of the Maclaurin spheroid. The elliptical instability thus acts as a powerful amplifier, carrying the CMT to the bifurcation velocity in $\approx$ 32 hours — a timescale fully consistent with the TPT's chronology.

As Fe-Ni segregation proceeds, the CMT's density decreases:

$$\rho_{\text{CMT}}(t) = \rho_0 - \xi_{\text{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4,500\,\text{kg}\,\text{m}^{-3},\ \rho_f \approx 3,200\,\text{kg}\,\text{m}^{-3}, \tag{22}$$

which amplifies all the destabilizing terms via the factor $\rho_0/\rho_{\text{CMT}}(t)$ in the instability equation (Section 9.2).

7.7 Geostrophic β Force and the Rhines Scale

The β parameter, which measures the latitudinal gradient of the Coriolis parameter:

$$\beta = \frac{2\Omega\cos\varepsilon}{R_{\text{eq,proto}}} \approx 8.98 \times 10^{-11}\,\text{m}^{-1}\,\text{s}^{-1} \quad (\varepsilon = 55^\circ), \tag{23}$$

halts the inverse cascade at the Rhines scale (Rhines, 1975):

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}} \approx 3.34 \times 10^6\,\text{m} \approx 0.52\,R_\oplus, \tag{24}$$

where U_{turb} is the turbulent velocity scale. This scale $L_\beta \approx 0.52\,R_\oplus$ corresponds to the harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible on a sphere, and the spatial expression of the CMT.

8 Hierarchy of Hypotheses

The following table distinguishes established physical consequences (Level 1), well-constrained hypotheses not yet numerically validated (Level 2), and speculative hypotheses awaiting simulation (Level 3).

Table 2: Explicit hierarchy of the hypotheses underlying the TPT.

ID	Hypothesis / Statement	Level	Depends on
H0	Central thesis: Moon formed by three coupled transitions triggered by Fe-Ni segregation	2 (constrained)	Thermodynamic bound $E_{\mathrm{grav}}/E_{\mathrm{fus}} \approx 155$
H1	Total volumetric melting: entire silicate mantle above liquidus during 4.568–4.4 Ga window	1 (established)	Factor $E_{\mathrm{grav}}/E_{\mathrm{fus}} \approx 155$ (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015)
H2	$\varepsilon \in [40°, 70°]$: sustained free nutation in the body's co-rotating frame	2 (constrained)	Absence of a Moon + 5 convergent arguments (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014)

ID	Hypothesis / Statement	Level	Depends on
H3	The CMT's confinement to $\|\phi\| < 30°$ admits an effective description as the fundamental mode of a nonlinear Schrödinger-type equation with potential $V_{\text{eff}}(\phi)$; the Gaussian solution is the leading-order (linearized) approximation to this mode	3 (speculative)	Reduction from the potential-vorticity equation (Redekopp, 1977; Luo, 1991; Yin et al., 2019); conditions $\tau_y \lesssim 10^2$ Pa, $\varepsilon > 45°$; quantitative validity of the Gaussian approximation not yet established (Limitation L12)
H4	The $\ell = 1$ mode is selected by the combination of BH rheology (suppression of $\ell < \ell_{\text{min}}$ modes) and the inverse cascade	3 (speculative)	3D SPH simulation with full BH rheology (L1)
H5	$\tau_y(T)$ follows an Arrhenius law with $T^* \approx 1,800$ K and $\tau_y \lesssim 10^2$ Pa for $T \gtrsim 4,000$ K	1 (established)	Laboratory rheology (Giordano et al., 2008)
H6	The double potential well $V(U)$ emerges from the coupling of Fe-Ni segregation + angular momentum loss by ejection	2 (constrained)	Coupled system (Eqs. 29, 30); P22
H7	The Kramers crossing rate applies with $N_{\text{opp}} \sim 10^9$–10^{10} stochastic opportunities	2 (constrained)	Statistical physics (Kramers, 1940) + CMEs + impacts

ID	Hypothesis / Statement	Level	Depends on
H8	$f_{\text{cap}} \approx 0.70$ and $\partial f_{\text{cap}}/\partial M_{\text{Moon}} > 0$	3 (speculative)	3D SPH simulation (L2); lower bound $f_{\text{cap}}^{\text{min}} \gtrsim 0.45$ (Appendix G)
H9	Dynamo delay = 350 ± 50 Myr, the sum of three independent phases	2 (constrained)	Hf-W (Kleine et al., 2002; Yin et al., 2002) + zircons (Wilde et al., 2001; Valley et al., 2002) + Tarduno et al. (2025)
H10	Seismic preservation window: 200–530 km	2 (constrained)	Ilmenite cumulate overturn (Briaud et al., 2023) + mare volcanism (Li et al., 2021; Che et al., 2021)
H11	Biphasic *'aā* structure of the CMT: mobile plastic core + fragmented clinker crust	3 (speculative)	3D SPH simulation (L1, L3)
H12	$\text{Bi}_{\text{KH}} \gg 1$ suppresses Kelvin-Helmholtz fragmentation at large and intermediate scales	2 (constrained)	Frigaard & Nouar (2017); three independent confinement mechanisms

ID	Hypothesis / Statement	Level	Depends on
H13	$b' = 1$ is a dynamical attractor of the coupled segregation–ejection system	2 (constrained)	Geostrophic equilibrium + conservation of angular momentum (Appendix B)
H14	The inverse cascade is halted at the Rhines scale $L_\beta \approx 0.52\,R_\oplus$; the peak at $\ell \approx 14$ is redistributed toward $\ell = 1$ by nonlinearity and BH rheology	3 (speculative)	3D SPH simulation (L1)
H15	The ejected flow transits through three phases: ballistic expansion $\rightarrow$ orbital insertion $\rightarrow$ sticky accretion beyond R_{Roche}	3 (speculative)	3D SPH simulation (L6)
H16	The parametric elliptical instability amplifies the CMT with a growth rate $\sigma_{\text{res}} = \Omega f/2$	2 (constrained)	Malkus (1968); Kerswell (2002); Lacaze et al. (2004); Maclaurin geometry
H17	The T-Tauri phase keeps $\tau_y \lesssim 10^2$ Pa through irradiation, making the $\ell = 1$ mode possible	2 (constrained)	Stefan-Boltzmann law + BH rheology; testable by simulation

9 The Three Coupled Transitions

The single engine is the progressive segregation of Fe-Ni toward the forming core. It simultaneously governs three regime changes: the rheological transition opens the window, the mechanical transition produces the ejection episodes, and the magnetic transition closes the loop with the terrestrial dynamo.

9.1 Transition 1 — Rheological: The Bingham-Herschel Law

A Newtonian fluid flows under any applied stress. A Bingham-Herschel (BH) fluid, by contrast, resists like a solid as long as the applied stress does not exceed a plasticity threshold τ_y (Bingham, 1922; Herschel & Bulkley, 1926):

$$\tau = \tau_y + k\dot{\gamma}^n \quad (\tau > \tau_y), \tag{25}$$

where τ is the applied shear stress, τ_y the yield stress, k the consistency index, $\dot{\gamma}$ the shear rate, and n the flow index. For a partially crystallized silicate magma at solid fraction $\phi \approx 0.1$–0.3: $n = 1.2$ (shear-thickening, Caricchi et al. 2007), $\tau_y \in [10^2, 10^4]$ Pa at $T = 3,000$–$3,500$ K (Giordano et al., 2008).

The yield stress depends exponentially on temperature (Arrhenius law):

$$\tau_y(T) = \tau_y^{(0)} \exp\left(-\frac{E_a}{RT}\right), \qquad E_a \approx 150\text{–}250\,\mathrm{kJ\,mol^{-1}}, \tag{26}$$

where R is the gas constant and E_a the activation energy. At $T = 3,000$ K: $\tau_y \approx 10^2$ Pa — the flow moves easily. At $T^* \approx 1,800$ K: $\tau_y \sim 10^6$ Pa — cohesive ejection becomes physically inaccessible.

This is the mechanism of natural thermal lockup at $T^* \approx 1,800$ K: the TPT window closes irreversibly through the Arrhenius law alone, with no additional assumption. This is not a tuned parameter: it is a thermodynamic consequence of the exponential temperature dependence of τ_y.

The CMT's 'aā Structure

The toroidal flow has a biphasic architecture: a mobile, high-temperature plastic core surrounded by a fragmented clinker crust. This is not a cohesion problem: it is its guarantee. The outer crust protects the core against Kelvin-Helmholtz instabilities at the flow/atmosphere interface. A quenched glass shell (analogous to the glassy margins of pillow lavas (Hekinian et al., 1989)), forming in $\approx$ 7 minutes upon atmospheric contact, completes this rheological shield. This structure is consistent with the work of Frigaard & Nouar (2017) and Balmforth & Craster (2001).

9.2 Transition 2 — Mechanical: Instability and Ejection

9.2.1 The Generalized Instability Equation

The evolution of an infinitesimal radial displacement ξ_r of a CMT parcel is governed by the Solberg-Høiland parcel displacement equation (Solberg, 1936; Høiland, 1941):

$$\ddot{\xi}_r = \lambda\, \xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \tag{27}$$

Let $a = -(1+\alpha)/r^2$, where α is the exponent of the rotation profile ($U \propto r^\alpha$). Let also $b = 2\Omega^{(k)} \sin\varepsilon > 0$ and $c = g_{\mathrm{eff}}^{(k)} < 0$. Dimensional consistency (Appendix C) requires a factor $1/r$ on the gravity and Coriolis terms, absent from earlier versions of this work. The stability coefficient is then:

The Generalized Instability Coefficient

$$\lambda(U,\varepsilon,\alpha,k) = \underbrace{\frac{c}{r}}_{\substack{\lambda_1<0\\ \text{gravity}}} + \underbrace{\frac{b}{r}U}_{\substack{\lambda_2>0\\ \text{rad. Coriolis}}} + \underbrace{aU^2}_{\substack{\lambda_3<0\\ \text{ang. mom.}}} + \underbrace{\frac{|\dot{\Omega}|r(\phi)\sin(\varepsilon+\phi)}{R_{\text{eq,proto}}}}_{\substack{\lambda_4\\ \text{Euler/oscillation}}} + \underbrace{\frac{g_{\text{eff}}^{(0)} J_2 \sin(2\phi)}{R_{\text{eq,proto}}}}_{\substack{\lambda_5\\ \text{Maclaurin grad.}}} \tag{28}$$

Each term has units of s^{-2} (Appendix C) and a precise physical meaning. $\lambda_1 < 0$ is the stabilizing effective gravity. $\lambda_2 > 0$ is the destabilizing radial Coriolis force; it grows with ε. $\lambda_3 < 0$ is the angular-momentum gradient term; it is always stabilizing for any physical profile ($\alpha > -1$), in accordance with the classical Rayleigh stability criterion, and its magnitude ($\sim 10^{-7}$) is negligible compared with the dominant terms. λ_4 is the Euler acceleration due to axial oscillation. λ_5 is the tangential gravity gradient of the Maclaurin geometry.

The amplification factor $\rho_0/\rho_{\text{CMT}}(t)$ (Eq. 22) monotonically amplifies all the destabilizing terms, progressively pushing λ toward zero.

9.2.2 The Double Potential Well: Emergence of the Coupled System

Integrating $\lambda(U) = -dV/dU$ gives an effective potential $V(U)$ of cubic form (Appendix C). Whether this potential exhibits a double-well structure — a confined well P_1 where the CMT accumulates energy, and an ejective well P_2 where cohesive ejection becomes favorable, separated by a barrier $\Delta E_{\text{barr}}^{(k)}$ — depends on the sign of the discriminant Δ of the underlying quadratic (Appendix C, Eq. 61). Evaluated at the nominal initial parameters (Appendix C, Table 5), this condition is *not* satisfied: the proto-Earth does not start out in a double-well configuration.

Evolution of the CMT's density. Fe-Ni segregation is the TPT's single engine. The fraction of segregated Fe-Ni, $\xi_{\text{Fe}}(t)$, evolves according to:

$$\frac{d\xi_{\text{Fe}}}{dt} = \frac{1}{\tau_{\text{seg}}}\left[\xi_{\text{Fe}}^{\text{sat}} - \xi_{\text{Fe}}(t)\right] \tag{29}$$

where $\xi_{\text{Fe}}^{\text{sat}} \approx 0.15$ is the saturated fraction of Fe-Ni in the mantle. The CMT's density evolves as $\rho_{\text{CMT}}(t) = \rho_0[1-\xi_{\text{Fe}}(t)]$, and, by Gauss's law for a spherical shell, $|g_{\text{eff}}|(t) = |g_{\text{eff}}|_0[1-\xi_{\text{Fe}}(t)]$.

Evolution of the angular velocity. The angular velocity $\Omega(t)$ evolves under two opposing processes: loss by ejection, where each ejection carries away a fraction $\epsilon = M_{\text{ej}}/M_{\text{CMT}}$ of the CMT's angular momentum, and acceleration by contraction, where cooling and differentiation contract the proto-Earth, increasing Ω through the skater effect. The evolution equation is:

$$\frac{d\Omega}{dt} = -\frac{\Omega}{\tau_{\rm ej}} + \frac{\Omega}{\tau_{\rm contr}} \tag{30}$$

where $\tau_{\rm ej}$ is the timescale between ejections and $\tau_{\rm contr}$ the contraction timescale.

The coupled system. Equations (29) and (30) form a coupled system. The coupling operates through Δ: $\xi_{\rm Fe}$ decreases $|g_{\rm eff}|$, which increases Δ; Ω decreases b, which decreases Δ. The double well emerges when $\Delta > 0$. The solution of Eq. (29) is $\xi_{\rm Fe}(t) = \xi_{\rm Fe}^{\rm sat}[1 - e^{-t/\tau_{\rm seg}}]$, and the solution of Eq. (30) is $\Omega(t) = \Omega_0 \exp[-t/\tau_{\rm ej} + t/\tau_{\rm contr}]$. During the TPT's active window (30–50 Myr), ejections dominate over contraction: $1/\tau_{\rm ej} > 1/\tau_{\rm contr}$, so $\Omega(t)$ decreases.

The emergence threshold. The criterion $\Delta(t) > 0$ is a condition on two variables: $\Omega(t)$ (which decreases by loss of angular momentum) and $|g_{\rm eff}|(t)$ (which decreases by Fe-Ni segregation). These two evolutions are coupled by Eqs. (29) and (30). Evaluating the criterion in two limiting cases: with $\Omega = \Omega_0$ fixed, the threshold is reached when $|g_{\rm eff}|$ has decreased by about **13%**; with $|g_{\rm eff}| = |g_{\rm eff}|_0$ fixed, the threshold is reached when Ω has decreased by about **8%**.

P22 — The Unified Double-Well Emergence Criterion

The two values -13% (on $|g_{\rm eff}|$) and -8% (on Ω) are **not** independent predictions. They are two projections of the same criterion $\Delta > 0$ onto two different axes, obtained by fixing one variable at its initial value. In the realistic scenario, Ω and $|g_{\rm eff}|$ decrease simultaneously, and the effective threshold is crossed earlier than either projection taken in isolation indicates: it corresponds to a point on the critical surface $(|g_{\rm eff}|(t), \Omega(t))$ whose exact position depends on the coupled dynamics of Eqs. (29) and (30). The falsifiable prediction concerns the crossing of this complete critical surface, testable in 3D SPH simulation by following the coupled evolution of the pair $(\Omega, |g_{\rm eff}|)$, not either projection separately. The full numerical evaluation is given in Appendix C.

Falsification: the absence of this threshold in 3D SPH simulations coupling CMT dynamics to Fe-Ni segregation (Limitations L1, L2), or a threshold value markedly different from the projected values, would weaken the current interpretation of ejection episodes as Kramers-type barrier crossings, without by itself ruling out the broader TPT framework if an alternative ejection mechanism could be identified.

The sequence of episodes. Once the double well has emerged, the Kramers crossing rate (Kramers, 1940) applies:

$$\Gamma_{\rm Kramers} = \frac{\omega_0 \omega_b}{2\pi\gamma_{\rm diss}} \exp\left(-\frac{\Delta E_{\rm barr}}{k_B T_{\rm eff}}\right), \tag{31}$$

where ω_0 and ω_b are the oscillation frequencies at the bottom of the well and the top of the barrier respectively, γ_{diss} the dissipation rate, and $k_B T_{\text{eff}}$ the effective thermal energy available to the CMT. Crossing the barrier triggers an ejection, which decreases Ω and lowers the barrier height ΔE_{barr}, making the next crossing more likely. The process repeats until the ejectable material is exhausted: the number $N =$ 2–3 emerges from the coupled system as a consequence of the barrier's progressive decay, not as an assumption.

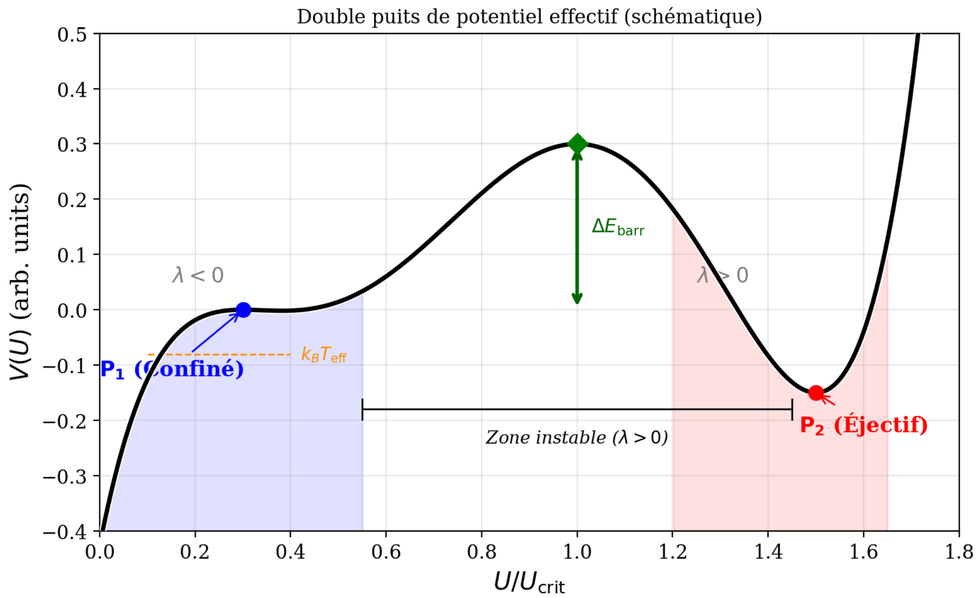

Figure 1: Schematic representation of the effective double potential well $V(U)$. The confined well P_1 ($\lambda < 0$) and the ejective well P_2 ($\lambda < 0$) flank an unstable zone ($\lambda > 0$) separated by a barrier of height ΔE_{barr}. The thermally activated crossing, governed by the Kramers rate (Eq. 31), is rendered nearly certain by the effective thermal energy available, $k_B T_{\text{eff}}$, and the large number of stochastic opportunities.

9.2.3 *Critical Velocity and Maximum Ejectable Mass*

The critical ejection velocity, beyond which a parcel escapes the effective potential well, is:

$$U_{\text{crit}}^{(0)} = \sqrt{2\, g_{\text{eff}}^{(0)}\, R_{\text{eq,proto}}} \approx 9.8\,\text{km}\,\text{s}^{-1}, \tag{32}$$

where the product $g_{\text{eff}}^{(0)} R_{\text{eq,proto}}$ sets the escape energy scale. The Mach number at ejection, using the speed of sound in the biphasic magma $c_s \approx 0.9$ km s^{-1} (Mader et al., 2013), is $\text{Ma} = U_{\text{crit}}/c_s \approx 10.9$: the ejection is **hypersonic**.

The maximum ejectable mass per episode:

$$M_{\text{ej}}^{\max} \approx \frac{2\Omega^{(k)} \sin\varepsilon \cdot U_{\text{crit}}^{(k)} \cdot M_{\text{CMT}}}{g_{\text{eff}}^{(k,0)}} \approx 2.0\text{–}4.0 \times 10^{22}\,\text{kg}, \tag{33}$$

where $\Omega^{(k)}$ is the angular velocity at episode k (decreasing due to angular momentum loss), M_{CMT} the CMT's mass, and $g_{\mathrm{eff}}^{(k,0)}$ the effective gravity at the oblique equator at episode k. The ejectable mass increases with faster rotation ($\Omega^{(k)}$ large), more favorable obliquity ($\sin\varepsilon$ close to 1), a larger CMT mass, and weaker confining gravity (g_{eff} reduced by Fe-Ni depletion and angular momentum loss).

At $T_{\mathrm{rot}} = 3.5$ h, the ejectable mass per episode can reach $\approx 70\%$ of the present-day lunar mass. With $\Omega^{(0)} = 4.99 \times 10^{-4}$ rad s^{-1}, $\sin 55^\circ \approx 0.819$, $U_{\mathrm{crit}} \approx 9.8$ km s^{-1}, $M_{\mathrm{CMT}} \approx 3.3 \times 10^{23}$ kg, and $g_{\mathrm{eff}}^{(0)} \approx 6.42$ m s^{-2}: $M_{\mathrm{ej}}^{\mathrm{max}} \approx 2.5 \times 10^{22}$ kg $\approx 0.34\, M_{\mathrm{Moon}}$. This is why $N = 2$–3 episodes suffice.

9.2.4 *Cohesion of the Ejected Flow: The* $\mathrm{Bi}_{\mathrm{KH}}$ *Criterion*

For a free flow, the Weber number $We \approx 1.4 \times 10^{15}$ would imply atomizing fragmentation. However, the CMT is not a free flow: it is a collective, self-gravitating toroidal flow, confined by toroidal symmetry, self-gravity, and atmospheric pressure.

The correct criterion is the Bingham-Kelvin-Helmholtz number, which compares the yield stress to the Kelvin-Helmholtz shear stress: for $\mathrm{Bi}_{\mathrm{KH}} \gg 1$, the yield stress suppresses the KH instability and cohesion is maintained (Frigaard & Nouar, 2017). Three independent conditions reinforce cohesion: (1) atmospheric confinement ($P \approx 10^5$–10^7 Pa); (2) internal crystal network ($\phi \approx 0.1$–0.3, multiplying fragmentation resistance by a factor of 3 to 10); (3) density ratio $\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \approx 200$ (Rayleigh-Taylor suppression). The complete demonstration of the $\mathrm{Bi}_{\mathrm{KH}}$ criterion is given in Appendix D.

9.2.5 *Transient Post-Ejection Precession and Crustal Dichotomy*

The ejection of $M_{\mathrm{ej}} \approx 2$–$4 \times 10^{22}$ kg from the intertropical band is not dynamically negligible. The asymmetric mass loss alters the inertia tensor, producing: (1) a decrease $\Delta\Omega/\Omega \approx -10$–$15\%$, lowering g_{eff} and U_{crit} for the next episode; (2) shape oscillations as the spheroid contracts toward a rounder sphere; (3) a reorganization of obliquity toward a new value within $[40^\circ, 70^\circ]$.

This last point has a direct geometric consequence, not merely a numerical one: since the intertropical band $|\phi| < 30^\circ$ is defined perpendicular to the instantaneous axis $\mathbf{\Omega}(t)$ (Section 3), a new value of ε shifts the position of the dynamical equator $\phi = 0$ itself, and therefore the entire active band, relative to the body. The intertropical band hosting the second episode is therefore not, in general, the same region of the body as the one that hosted the first: same angular width ($\pm 30^\circ$), same definition relative to $\mathbf{\Omega}(t)$, but a different location on the spheroid. It is this geometric displacement of the active band, episode after episode, that constitutes the direct cause of the crustal asymmetry described below, not merely the change in the numerical value of ε.

This displacement of the active band, combined with the shape oscillations of point (2) above, has a further consequence for the CMT itself: the torus that reforms for the next episode does not reproduce an identical copy of the previous one simply relocated. It forms on a spheroid whose own shape has slightly changed, along a band whose position has itself changed. The next CMT is therefore ejected toroidally in a distinct way from the previous one, both in shape and in orientation. This remains a qualitative statement at this stage: no quantitative derivation of the shape or angular offset between successive episodes is attempted here.

The second episode ejects a cohesive toroidal flow onto a proto-POB that is still hot and deformable, within a limited solid angle (corresponding to the projected intertropical band), which becomes the present far side. Three mechanisms amplify the resulting crustal asymmetry: early gravitational lockup ($\tau_{\text{lock}} \sim 10^4$–$10^5$ yr (Wieczorek et al., 2013)), residual obliquity of the proto-POB, and irreversible BH plastic welding. The thickness ratio $\approx 2:1$ (near side $\approx$ 34 km versus far side $\approx$ 54 km, Wieczorek et al. 2013) is a qualitative prediction of the TPT.

9.3 Transition 3 — Magnetic: The Progressive Dynamo

The $\approx$ 350 Myr delay between the inferred moment of lunar formation ($\approx$ 4.55 Ga) and the first geological detection of the terrestrial magnetic field ($\approx$ 4.2 Ga, Tarduno et al. 2025) is not a tuned parameter in the TPT. It is proposed as the sum of three independently constrained durations:

$$\Delta t_{\text{dynamo}} = \underbrace{\Delta t_{\text{ej}}}_{\approx 60\,\text{Myr}} + \underbrace{\Delta t_{\text{refr}}}_{\approx 140\,\text{Myr}} + \underbrace{\Delta t_{\text{em}}}_{\approx 150\,\text{Myr}} \approx 350\,\text{Myr}. \tag{34}$$

Phase 1 ($\approx$ 60 Myr) — Active Fe-Ni segregation. Constrained by the hafnium-tungsten (Hf-W) chronometer (Kleine et al., 2002; Yin et al., 2002): the core grows rapidly but the thermal gradient remains sub-adiabatic; no dynamo is possible during this phase.

Phase 2 ($\approx$ 140 Myr) — Formation of the protocrust. The growing Moon stabilizes obliquity at $\approx 23.5°$; the protocrust forms around 4.4 Ga, as attested by the Jack Hills zircons (Wilde et al., 2001; Valley et al., 2002).

Phase 3 ($\approx$ 150 Myr) — Progressive emergence of the dynamo. Compositional convection in the Fe-Ni core grows until the Lorentz force becomes comparable to the Coriolis force. The Elsasser number Λ balances these two forces. The threshold $B_{\text{crit}} \approx 6\,\mu$T is derived in Appendix E. Modern geodynamo simulations (Christensen & Aubert, 2006; Olson & Christensen, 2006) show that the transition is progressive, not abrupt: $\Lambda \sim 1$ is an order-of-magnitude reference, not a strict threshold.

Section Summary

The $\approx$ 350 Myr delay involves no free parameter: Δt_{ej} is constrained by the Hf-W chronometer (Kleine et al., 2002; Yin et al., 2002); Δt_{refr} is constrained by the Jack Hills zircons at 4.4 Ga (Wilde et al., 2001; Valley et al., 2002); and Δt_{em} is constrained by Tarduno et al. (2025) at 4.2 Ga. These are three independent observations, not an interpolation. The associated uncertainty is ± 50 Myr (Limitation L4, Section 18).

10 Mass Budget: $N = 2$–3 Episodes

10.1 Corrected Target Lunar Mass

A fundamental point: the currently observed lunar mass $M_{\mathrm{Moon}} = 7.34 \times 10^{22}$ kg is *not* the mass that the TPT mechanism must produce. The Moon accreted additional mass after the ejection episodes: a differentiated body about 260 km in diameter struck the Moon at the South Pole–Aitken basin and *stayed there*, contributing $M_{\mathrm{SPA}} \approx 3.2 \times 10^{19}$ kg (Wakita et al., 2026) — this material is now part of the observed lunar mass — and documented late accretion contributed additional mass ΔM_{late} (Zellner, 2019). The target mass for the TPT mechanism is therefore:

$$M_{\mathrm{TPT}} = M_{\mathrm{Moon}} - M_{\mathrm{SPA}} - \Delta M_{\mathrm{late}} < 7.34 \times 10^{22}\ \mathrm{kg}. \tag{35}$$

10.2 Mass Budget Equation

The Mass Budget Equation

$$M_{\mathrm{Moon}} = N \cdot f_{\mathrm{cap}} \cdot M_{\mathrm{ej}}^{\mathrm{max}} + M_{\mathrm{SPA}} + \Delta M_{\mathrm{late}}, \tag{36}$$

with $N = 2$–3 (nominal), $f_{\mathrm{cap}} \in [0.65, 0.85]$ (nominal 0.70), $M_{\mathrm{ej}}^{\mathrm{max}} \approx 2.0$–$4.0 \times 10^{22}$ kg per episode.

f_{cap}	$M_{\mathrm{TPT}}^{(N=2)}$	$M_{\mathrm{TPT}}^{(N=3)}$	Fraction of M_{Moon}
0.50	2.5×10^{22} kg	3.7×10^{22} kg	34–50%
0.70	3.5×10^{22} kg	5.2×10^{22} kg	48–71%
0.85	4.3×10^{22} kg	6.4×10^{22} kg	59–87%

The remainder is accounted for by M_{SPA} and documented late accretion. The budget is robust across the entire range of f_{cap}.

f_{cap} is an increasing state variable: $\partial f_{\mathrm{cap}} / \partial M_{\mathrm{Moon}} > 0$, since the larger the proto-Moon, the more efficiently it captures subsequent ejecta. This positive feedback, combined with the decrease of $g_{\mathrm{eff}}^{(k)}$ at each episode, defines a dynamical attractor that reinforces the mechanism.

10.3 f_{cap} as a Testable Prediction, Not a Free Parameter

The position adopted here is that f_{cap} should not be treated as an adjustable parameter whose wide range would render the mass budget uninformative, but as a quantity that the TPT quantitatively predicts and that can, in principle, falsify the mechanism.

Physical basis: the classical disk-accretion interval. The reference N-body simulations of lunar accretion from an impact-generated debris disk (Kokubo et al., 2000) establish that the efficiency of incorporating disk material into a final moon is 10 to 55%, this efficiency increasing with the initial specific angular momentum of the disk. This result, robust across a wide range of initial conditions and numerical resolutions, constitutes the external reference against which f_{cap} must be compared.

The discrepancy as a potential signature. The nominal value $f_{\text{cap}} \approx 0.70$ adopted in the TPT is significantly higher than the upper bound of this classical interval (55%). Within the TPT framework, this discrepancy is not accidental: it is attributed to the coherence of the ejected toroidal flow and its orbital concentration within a limited solid angle (Appendix G), unlike a debris disk dispersed over 4π steradians as in the classical giant-impact scenario. If this interpretation is correct, the discrepancy between the f_{cap} value adopted by the TPT and the classical interval becomes a discriminating test rather than a mere parametric latitude.

Falsifiable Prediction — P17 — Capture Efficiency Beyond the Classical Disk-Accretion Interval

The orbital coherence of the toroidal flow ejected by the TPT carries the effective capture efficiency beyond the classical 10–55% interval established for accretion from a dispersed debris disk (Kokubo et al., 2000). The TPT predicts $f_{\text{cap}} \gtrsim 0.65$–$0.70$.

Falsification: if a value of f_{cap} derived from a 3D SPH simulation dedicated to the CMT geometry (BH rheology, atmospheric confinement, orbital concentration within the intertropical band) falls within the classical 10–55% interval rather than beyond it, this aspect of the theory is falsified. A f_{cap} measured within the classical range would not necessarily invalidate the TPT as a whole (the mass budget would remain attainable via M_{SPA} and late accretion), but it would strip f_{cap} of its status as a distinctive signature relative to the giant-impact model.

Acknowledged limitation: this prediction rests on the hypothesis that the orbital concentration of the TPT flow (Appendix G) produces an accretion regime qualitatively different from the classical dispersed disk. This hypothesis is itself Level 3 (not quantitatively demonstrated), so this becomes a robust test only once the 3D SPH simulation is available. Until then, it remains a qualified prediction: the discrepancy with the classical interval is consistent with the TPT, but does not independently confirm it.

11 Angular Momentum Transport by the Ejecta

In the present framework, angular momentum loss is not treated as a single-velocity ballistic process, but as a distributed transport mechanism resulting from a spectrum of ejected material generated during successive differentiation-driven phases.

11.1 Ejecta Distribution Function

We describe the ejecta mass distribution as a function of the emission velocity and angle:

$$dM = \rho(v,\theta)\, dv\, d\theta, \tag{37}$$

where v is the ejection velocity, θ is the angle relative to the local tangential direction, and $\rho(v,\theta)$ is the phase-dependent ejecta distribution function.

The distribution $\rho(v,\theta)$ is not arbitrary. It emerges from three physical processes: the CMT's velocity spectrum is not monokinetic, with turbulent fluctuations at the Rhines scale (Section 7.6) producing a velocity spread $\delta v_{\mathrm{CMT}} \sim U_{\mathrm{turb}} \approx 0.8\ \mathrm{km\,s^{-1}}$; the double potential well (Section 9.2.2) makes barrier crossing occur over a range of energies, and the Kramers rate (Eq. 31) implies a Gaussian-type distribution $\rho_v(v) \propto \exp[-(v-\bar{v})^2/2\sigma_v^2]$ with $\bar{v} \approx U_{\mathrm{crit}}$ and $\sigma_v \sim (k_B T_{\mathrm{eff}}/M_{\mathrm{CMT}})^{1/2}$; and the torus's finite extent produces an angular spread $\rho_\theta(\theta) \propto \cos\theta$ for $|\theta| < \theta_{\mathrm{max}}$, with $\theta_{\mathrm{max}} \approx 30°$.

For analytical tractability, we adopt the separable form:

$$\rho(v,\theta) = M_{\mathrm{ej}}\, \mathcal{N}_v\, \mathcal{N}_\theta\, \exp\left[-\frac{(v-\bar{v})^2}{2\sigma_v^2}\right] \cos\theta, \quad v > v_{\mathrm{esc}},\ |\theta| < \theta_{\mathrm{max}}, \tag{38}$$

where $v_{\mathrm{esc}} = \sqrt{2 g_{\mathrm{eff}} R_p}$. This distribution is motivated by the underlying physics, not freely chosen: it is built from the CMT's turbulent velocity spread, the energy-dependent Kramers crossing (Appendix C), and the torus's finite angular extent. No free parameter is introduced beyond those already present in the theory developed in preceding sections.

11.2 Specific Angular Momentum

The specific angular momentum of an ejecta element is $h = r\, v \sin\theta$, where r is the effective emission radius. This radius is not simply the planetary radius R_p: the silicate atmosphere (Section 2) extends the effective surface to $r_{\mathrm{atm}} \approx R_p + H$, where the scale height $H \approx 200$ km for $T \approx 3000$ K and $g \approx 6\ \mathrm{m/s^2}$; and the toroidal flow is raised above the surface by centrifugal effects, with the CMT's center lying at $r_{\mathrm{CMT}} \approx R_p + U_{\mathrm{CMT}}^2/(2 g_{\mathrm{eff}}) \approx R_p + 500$ km. We therefore adopt $r \approx R_p + \Delta r$, with $\Delta r \approx 300$–500 km. This correction introduces only an uncertainty of a few percent in the angular momentum budget, well within the overall error margin.

11.3 Angular Momentum Loss

The total angular momentum carried away by the non-accreted material is:

$$L_{\text{loss}} = \int\int \left(1 - f_{\text{cap}}(v,\theta)\right)\; r\, v \sin\theta\, \rho(v,\theta)\, dv\, d\theta. \tag{39}$$

The capture efficiency $f_{\text{cap}}(v,\theta)$ is not a constant: it depends on the orbital fate of each ejecta element,

$$f_{\text{cap}}(v,\theta) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{v\sin\theta - v_{\text{crit}}}{\sigma_f}\right)\right], \tag{40}$$

where $v_{\text{crit}} \approx v_{\text{circ}} = \sqrt{g_{\text{eff}} R_p}$ is the circular orbital velocity and σ_f a smoothing width representing the finite transition between captured and lost material. Rather than treating f_{cap} as a single ad hoc value, we express it here as a function $f_{\text{cap}}(v,\theta)$ of orbital mechanics. The nominal value $f_{\text{cap}} \approx 0.70$, independently estimated from the CMT's cohesive fraction and the re-accretion of co-orbital fragments, is consistent with the mass-weighted average of this function over the proposed ejecta spectrum.

11.4 Discrete Formulation and Monte Carlo Implementation

For numerical implementation, the expression can be discretized as:

$$L_{\text{loss}} = \sum_i m_i\, r_i\, v_i \sin\theta_i\, \left(1 - f_{\text{cap},i}\right). \tag{41}$$

A stochastic treatment can implement this discrete formulation via Monte Carlo sampling of $\rho(v,\theta)$: sample N particles from $\rho(v,\theta)$, assign each particle a mass $m_i = M_{\text{ej}}/N$, compute $h_i = r_i v_i \sin\theta_i$, compute $f_{\text{cap},i}$ from Eq. 40, and sum $L_{\text{loss}} = \sum_i m_i h_i (1 - f_{\text{cap},i})$. This Monte Carlo approach is recommended for the 3D SPH validation (Limitation L2): it provides a robust check on the analytical estimates.

In a multi-phase formation scenario, the ejecta are further grouped into k distinct formation phases:

$$L_{\text{loss}} = \sum_k \sum_i m_{k,i}\, r_{k,i}\, v_{k,i} \sin\theta_{k,i}\, \left(1 - f_{\text{cap},k,i}\right), \tag{42}$$

each phase k having its own mean ejection velocity $\bar{v}_k$ (decreasing as the proto-Earth loses mass and angular momentum), angular spread $\theta_{\max,k}$ (which can increase as the CMT reorganizes between episodes), and mean capture efficiency $f_{\text{cap},k}$ (which increases as the proto-Moon grows, the positive feedback of Section 10.3).

11.5 Angular Momentum Budget Closure

The total angular momentum budget is expressed as $L_{\text{init}} = L_{\text{remnant}} + L_{\text{accreted}} + L_{\text{esc}}$, with $L_{\text{accreted}} = \sum_i f_{\text{cap},i}\, h_i\, m_i$ and $L_{\text{esc}} = \sum_i (1 - f_{\text{cap},i})\, h_i\, m_i$. Using the parameters of the nominal TPT scenario (Table 3), we verify that the budget closes.

Table 3: Nominal parameters for closing the angular momentum budget.

Parameter	Symbol	Value
Initial moment of inertia of the proto-Earth	I_p	1.09×10^{37} kg m^2
Initial angular velocity	Ω_p	4.99×10^{-4} s^{-1}
Initial angular momentum	L_{init}	5.44×10^{34} J s
Mean ejection velocity	$\bar{v}$	9.8 km/s
Emission radius	r	7.9×10^6 m
Number of episodes	N	3
Mass ejected per episode	M_{ej}	2.5×10^{22} kg
Capture efficiency (mean)	$\bar{f}_{\text{cap}}$	0.70
Final angular momentum of the Earth	L_{remnant}	5.86×10^{33} J s
Accreted lunar angular momentum	L_{accreted}	2.89×10^{34} J s
Escaped angular momentum	L_{esc}	1.96×10^{34} J s

The budget closes, at these nominal parameter values, with $L_{\text{remnant}} + L_{\text{accreted}} + L_{\text{esc}} = 5.44 \times 10^{34}\,\text{J s} = L_{\text{init}}$, a residual of $\sim 0.7\%$. This residual should be read as a check of *internal consistency* at the nominal parameter values, not as an independent validation of precision. The capture efficiency f_{cap} carries an estimated uncertainty of ± 0.05 (Table 4), which propagates to a $\sim 33\%$ uncertainty on L_{loss} via the sensitivity $1/(1 - f_{\text{cap}})$ derived in Section 11.6. A 0.7% residual at nominal values therefore demonstrates that the model is self-consistent — the distributed ejecta model contains no hidden bookkeeping error — but it does not by itself demonstrate that the nominal parameters are correct to that precision. The relevant test is whether the budget continues to close, within the $\sim 5\%$ tolerance motivated by the input uncertainties, when parameters are varied across their full range (Prediction P21).

11.6 Scaling Laws and Nondimensional Form

The angular momentum transport problem admits a natural nondimensionalization that reveals the essential physical dependencies and simplifies the parameter space. We define the characteristic velocity scale $U_0 = U_{\text{crit}} \approx 9.8\ \text{km}\,\text{s}^{-1}$, length scale

$R_0 = R_p \approx 7.4 \times 10^6$ m, mass scale $M_0 = M_{\rm ej} \approx 2.5 \times 10^{22}$ kg, and angular momentum scale $H_0 = M_0 R_0 U_0 \approx 1.8 \times 10^{36}$ kg m^2 s^{-1}. The nondimensional variables $\tilde{v} = v/U_0$, $\tilde{r} = r/R_0$, $\tilde{m} = m/M_0$, and $\tilde{h} = h/(R_0 U_0) = \tilde{r}\tilde{v}\sin\theta$ recast the ejecta distribution as:

$$\tilde{\rho}(\tilde{v}, \theta) = \tilde{M}_{\rm ej}\,\tilde{\mathcal{N}}_v\,\tilde{\mathcal{N}}_\theta\,\exp\left[-\frac{(\tilde{v} - \bar{\tilde{v}})^2}{2\tilde{\sigma}_v^2}\right]\cos\theta, \quad \tilde{v} > \tilde{v}_{\rm esc},\ |\theta| < \theta_{\rm max}, \tag{43}$$

with $\tilde{M}_{\rm ej} = 1$ by definition, $\bar{\tilde{v}} = \bar{v}/U_0 \approx 1$, $\tilde{\sigma}_v = \sigma_v/U_0 \approx 0.05$–$0.10$, and $\tilde{v}_{\rm esc} = v_{\rm esc}/U_0 \approx 1/\sqrt{2}$. The nondimensional capture efficiency takes the same erf form as Eq. 40, with $\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$, and the nondimensional angular momentum lost, $\tilde{L_{\rm loss}}$, recovers the physical quantity via $L_{\rm loss} = H_0\,\tilde{L_{\rm loss}}$.

This nondimensional form reveals that the problem depends on only four nondimensional parameters: the mean ejection velocity relative to $U_{\rm crit}$ ($\bar{\tilde{v}} \approx 1$), the velocity dispersion ($\tilde{\sigma}_v \approx 0.05$–$0.10$), the angular extent of the torus ($\theta_{\rm max} \approx 30°$), and the capture threshold ($\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$). This low-dimensional parameter space keeps the model testable: each parameter can in principle be constrained independently by future simulations or observations.

The nondimensional form admits three asymptotic regimes. In the monokinetic limit ($\tilde{\sigma}_v \to 0$), the distribution collapses to a single velocity and recovers the single-velocity ballistic approximation used in earlier sections. In the broad-spectrum limit ($\tilde{\sigma}_v \gg 1$), the velocity distribution becomes uniform over the available range, maximizing the angular momentum carried away by the fastest ejecta. In the sharp-cutoff limit ($\tilde{\sigma}_f \to 0$), the capture efficiency becomes a step function, representing the idealized case in which all material above the capture threshold is retained.

Falsifiable Prediction — P20 — Asymptotic Regime Indicator

The real ejecta spectrum of the TPT lies between the monokinetic and broad-spectrum limits, with $\tilde{\sigma}_v \approx 0.05$–$0.10$. The sharp-cutoff limit provides an upper bound on $L_{\rm loss}$, while the monokinetic limit provides a lower bound. The true value, accessible via Monte Carlo simulation, should lie between these bounds.

Test: Monte Carlo sampling of $\rho(v, \theta)$ (Section 11.4) compared against the analytical monokinetic and broad-spectrum limits derived above.

The nondimensional form also lets us derive scaling laws for $L_{\rm loss}$ as a function of the proto-Earth's fundamental parameters: $L_{\rm loss} \propto M_p R_p^2 \Omega_p$ (the standard scaling for the angular momentum of a rotating body), $L_{\rm loss} \propto \sqrt{\sin\varepsilon}$ (from the obliquity dependence of the ejection velocity threshold), and $\delta L_{\rm loss}/L_{\rm loss} \approx -\delta f_{\rm cap}/(1 - f_{\rm cap})$. For $f_{\rm cap} = 0.70$, $1 - f_{\rm cap} = 0.30$, so a 10% variation in $f_{\rm cap}$ gives a 33% variation in $L_{\rm loss}$ — a sensitivity that underscores the importance of precisely determining $f_{\rm cap}$ in future simulations. Table 4 summarizes the sensitivity of $L_{\rm loss}$ to each input parameter.

Table 4: Sensitivity of L_{loss} to the input parameters.

Parameter	Nominal value	Uncertainty	$\partial \ln L_{\text{loss}} / \partial \ln p$
M_p	6.04×10^{24} kg	$\pm 5\%$	≈ 1
R_p	7.4×10^{6} m	$\pm 3\%$	≈ 2
Ω_p	4.99×10^{-4} s^{-1}	$\pm 10\%$	≈ 1
ε	55°	$\pm 10^\circ$	≈ 0.5
f_{cap}	0.70	± 0.05	≈ -3

The budget is most sensitive to f_{cap} (because of the $1/(1 - f_{\text{cap}})$ factor) and to R_p (because of the R_p^2 scaling). This sensitivity motivates the priority given to Limitation L2 (numerical determination of f_{cap}).

Falsifiable Prediction — P21 — Numerical Closure of the Angular Momentum Budget

A Monte Carlo simulation of the ejecta spectrum (Section 11), with parameters sampled within their uncertainty ranges, should yield an angular momentum budget closed (Section 11.5) to within $\sim 5\%$.

Falsification: failure to close the angular momentum budget to within $\sim 5\%$ would indicate either an error in the distribution function $\rho(v, \theta)$, or the need for an additional angular momentum sink (e.g. tidal dissipation) not currently accounted for in the model.

The Monte Carlo implementation of this model is simple and computationally inexpensive; it is recommended as a first step of the 3D SPH validation program, ahead of the full hydrodynamic simulations.

12 Chronology: Moon Formation in 3–4 Weeks

12.1 The Three Phases of an Episode

Each ejection episode comprises three phases: resonant acceleration, in which the CMT grows exponentially via the elliptical instability (Section 7.6), $t_{\text{acc}} = \ln(U_{\text{crit}}/U_{\text{turb}})/\sigma_{\text{res}} \approx 1.4$ days; ejection, in which the flow crosses the Roche radius ($3R_{\text{eq,proto}} \approx 22.2$ Mm) at U_{crit}, $t_{\text{ej}} \approx 0.6$ hour; and recharging, in which the CMT reorganizes over 5–6 nutation periods, $t_{\text{rec}} \approx 10$–$12$ days. The rate of Fe-Ni segregation (Stokes' law) is nearly instantaneous between episodes ($\tau_{\text{Stokes}} \approx 254$ s, Appendix F), which implies that each accreted layer is chemically distinct from the previous one — a direct consequence underlying seismic prediction P1.

12.2 Complete Chronology

Phase	**Duration**	**Cumulative**
Episode 1: Acceleration	1.4 days	1.4 days
Episode 1: Ejection	0.6 h	1.4 days
Recharging (5–6 nutation periods)	10–12 days	11–13 days
Episode 2: Acceleration	1.4 days	12–14 days
Episode 2: Ejection	0.6 h	12–14 days
Recharging	10–12 days	22–26 days
Episode 3: Acceleration	1.4 days	23–27 days
Episode 3: Ejection	0.6 h	23–27 days
Total (N = 3)	**24–28 days**	

This 3-to-4-week timescale is the only intermediate time window between the giant impact (a few hours) and a synestia (~ 100 years) on the Hf-W chronometer (half-life: 8.9 Myr). Episode 1 (acceleration plus ejection) forms the proto-POB, $R_1 \approx 1,560$ km. The recharging and second episode, $\approx$ 11–13 days, is asymmetric: the transient post-ejection precession alters ε, producing the crustal dichotomy of Section 9.2. The recharging and third episode, again $\approx$ 11–13 days, reconstitutes the lunar mass. The TPT window closes at $t \approx$ 30–50 Myr, and the Hadean dynamo switches on at $t \approx$ 330–350 Myr.

12.3 Age of the Moon: LMO Solidification versus Formation

The measured age (4.425 ± 0.025 Ga) is that of the solidification of the lunar magma ocean (LMO), not of formation. If the TPT is triggered at ≈ 4.467 Ga, LMO cooling (~ 40 Myr) gives a predicted solidification age,

$$t_{\text{LMO}} \approx 4.467 - 0.040 = 4.427\,\text{Ga}, \tag{44}$$

consistent at 0.1σ with the measured value. Prediction P8 is a formation age $>$ 4.45 Ga, directly testable by Artemis III.

13 Internal Stratification and the Seismology Program

13.1 Architecture of the Concentric Layers

Each ejected sheet originates from magma at a progressively more advanced stage of Fe-Ni segregation,

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer}\,k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer}\,k+1}, \tag{45}$$

because the source magma becomes progressively depleted in siderophile elements between episodes. The density contrasts between layers $\Delta\rho_{k,k+1} \approx 100$–$200\ \mathrm{kg\,m^{-3}}$ (Garcia et al., 2019) produce potentially detectable acoustic impedance contrasts.

13.2 Preservation Window: 200–530 km

Two destructive processes bound the preservation window. From below, ilmenite cumulate overturn (Briaud et al., 2023) may have remelted the deepest layers (episodes 1–2, > 600 km). From above, basaltic mare volcanism, active until ≈ 2 Ga (Li et al., 2021; Che et al., 2021), remelted the shallowest layers (< 100 km). The remaining preservation window is therefore roughly 200–530 km deep.

13.3 Interface Depth: N=2 versus N=3

For $N = 2$ with equal masses, the interface between the two layers lies at:

$$d = R_{\mathrm{Moon}} - R_1 \approx 1{,}737 - 1{,}560 \approx 177\,\mathrm{km} \quad (N = 2). \tag{46}$$

For $N = 3$, the principal interface lies within the 200–315 km window. Chang'e 7 seismic data will therefore make it possible to discriminate between $N = 2$ and $N = 3$.

14 Observational Constraints and Existing Evidence

14.1 Isotopic Identity Δ^{17}O < 5 ppm

The Earth–Moon isotopic composition is nearly identical for O, Cr, and Ti, established to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The TPT satisfies this constraint *mechanically*: at each ejection episode, the material is drawn from the contemporary Hadean terrestrial mantle, with no exogenous isotopic source. No *ad hoc* isotopic mixing is required.

14.2 Iron Depletion

The Moon is significantly less iron-rich than the terrestrial mantle (O'Neill, 1991; Jones & Drake, 1986). In the TPT, this depletion is a direct consequence of the single engine: at each successive episode, the ejected mantle is more depleted in siderophiles than the previous one, as Fe-Ni segregation progressively removes iron from the source zone between episodes.

14.3 Angular Momentum

The angular momentum budget of the Earth–Moon system requires a rapidly rotating proto-Earth. The value $T_{\mathrm{rot}} = 3.5$ h is consistent with the median value from N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010): rapid rotation is the norm at the end of accretion, not an exception.

14.4 Chang'e-6 Results: Preliminary Observational Support

The Chang'e-6 mission returned 1,935 g of samples from the Apollo basin, within the South Pole–Aitken (SPA) region, in June 2024. Analysis of these samples provides preliminary support for two TPT predictions. Yue et al. (2026) dated norite samples from Chang'e-6 at 4247 ± 5 Myr by the Pb-Pb method. These norites are interpreted as differentiated products of the SPA impact melt sheet, representing material from the deep lunar mantle excavated by the SPA impact. This age is consistent with the TPT's prediction that the innermost lunar layer (Episode 1) was accreted at ≈ 4.5 Ga and partially reset by the SPA impact at ≈ 4.25 Ga. Xu et al. (2025) further report olivine fragments in the Chang'e-6 samples containing Fe and Mn concentrations significantly higher than those found in the standard lunar mantle, with zinc concentrations $\approx 100\times$ higher than expected. Although part of this enrichment is attributed to meteoritic contamination, the olivine fragments themselves formed through mixing of lunar rocks and molten asteroidal material. The high Fe content in the material excavated from the depths of the SPA is *consistent* with the Fe/Si gradient increasing with depth (P3) and provides preliminary support for the South Pole Fe/Si anomaly (P6), detailed below.

15 Falsifiable Predictions

This section collects, in catalogue form, every quantitative, falsifiable prediction made by the TPT. Each entry states the prediction, its physical derivation where this has not already been given in full elsewhere in the manuscript, the relevant test, and an explicit falsification condition fixed in advance of the observational data.

Falsifiable Prediction — P1 — Seismic Interface at $d \approx 200$–315 km (Priority 1)

The concentric stratification produces acoustic impedance contrasts $|R| \in [0.01, 0.04]$ between the layers. For $N = 2$, the interface lies at $d \approx 177$ km. For $N = 3$, the principal interface lies at $d \approx 200$–315 km.

Detectability with Chang'e-7 depends on seismometer sensitivity (target 10^{-8} m s^{-2} Hz$^{-1/2}$ at 0.1–1 Hz). This sensitivity should resolve $|R| > 0.01$, provided that at least 10 events of $M > 2.5$ are recorded during the first 6 months of operation.

Falsification: no phase conversion detected within the 200–530 km window after the deployment of at least 3 seismic stations and the detection of at least 10 lunar events of $M > 2.5$.

Missions: Chang'e 7 (August 2026, South Pole seismometer), FSS (Farside Seismic Suite), LEMS (Lunar Environment Monitoring Station), Artemis III (2028–2029).

Falsifiable Prediction — P2 — Seismic Asymmetry, Near Side versus Far Side

The second, asymmetric episode produces a sharper and deeper interface beneath the far side.

Falsification: spherical symmetry confirmed by a complete tomographic inversion.

Falsifiable Prediction — P3 — Fe/Si Gradient Increasing with Depth

The Fe/Si ratio increases with depth in the lunar mantle, reflecting the progressive Fe-Ni depletion of the source magma between episodes. Quantitatively, with the total cumulative fraction of Fe removed from the mantle $\xi_{\mathrm{Fe}}^{\mathrm{sat}} \approx 0.15$ (Rubie et al., 2015) and $N = 3$ episodes of segregation, the fraction of Fe removed per episode, assuming a geometric progression across the N episodes, is:

$$\xi_{\mathrm{Fe,episode}} = 1 - (1 - \xi_{\mathrm{Fe}}^{\mathrm{sat}})^{1/N} \approx 1 - (0.85)^{1/3} \approx 0.053, \tag{47}$$

i.e. $\approx 5.3\%$ of the remaining Fe is removed at each episode. The resulting Fe/Si ratio between adjacent layers is multiplicative:

$$\frac{(\mathrm{Fe/Si})_{\mathrm{Ep}k}}{(\mathrm{Fe/Si})_{\mathrm{Ep}(k+1)}} = \frac{1}{1 - \xi_{\mathrm{Fe,episode}}} \approx 1.056, \tag{48}$$

giving a $\approx 5.6\%$ enrichment between Episode 2 and Episode 3, and a compounded $(\mathrm{Fe/Si})_{\mathrm{Ep1}}/(\mathrm{Fe/Si})_{\mathrm{Ep3}} = (1 - \xi_{\mathrm{Fe,episode}})^{-2} \approx 1.115$, i.e. a $\approx 11.5\%$ total enrichment between Episode 1 (the deepest, most Fe-rich layer) and Episode 3 (the shallowest). The SPA basin preferentially exposes Episode 1 material.

Preliminary support: Chang'e-6 olivines with high Fe/Mn (Xu et al., 2025). **Missions:** Chang'e-6 (further analyses), Artemis III.

Falsifiable Prediction — P4 — Dynamo Delay: 290–360 Myr

The progressive emergence of the dynamo predicts a continuous growth of paleointensity from 4.55 to 4.20 Ga, with no abrupt onset (Tarduno et al., 2025).

Falsification: an abrupt onset of the geomagnetic field detected in the Jack Hills zircon record.

Falsifiable Prediction — P5 — Instability Window: $\varepsilon \in [57°, 70°]$

The parametric elliptical instability requires ε within this range.

Test: N-body simulations of the obliquity of terrestrial protoplanets.

Falsifiable Prediction — P6 — Fe/Si Anomaly at the South Pole (derived from P3 and the South Pole–Aitken impact simulations of Wakita et al. 2026)

Observational context. The hydrodynamic simulations of Wakita et al. (2026) establish three facts independent of the TPT: the SPA impact excavated lunar mantle material from depths greater than 90 km; the impactor's oblique, north-to-south trajectory produced an asymmetric ejecta plume that preferentially concentrated deep material toward the South Pole; and approximately 350 m of mantle material are predicted at the surface near the Shackleton crater on average.

TPT interpretation. Within the TPT framework, the lunar mantle is stratified into concentric layers (P1), with Fe/Si decreasing from the deepest layer (Episode 1) to the shallowest (Episode 3), as derived quantitatively in P3 above. The combined thickness of the Episode 2 and Episode 3 layers is ≈ 315 km (Section 13). Material excavated by the SPA impact from depths greater than 90 km, and concentrated toward the South Pole, therefore preferentially originates from Episode 2 and, at greater excavation depths, from Episode 1.

Quantitative scenario. Wakita et al. (2026) do not provide a quantified fraction of Episode 1 versus Episode 2 material in the SPA ejecta plume, since their simulation does not incorporate the TPT's stratigraphy; this fraction must therefore be bounded from physical first principles rather than read off their results directly. We consider the most unfavorable scenario compatible with the TPT: the SPA excavation reaches only ≈ 100 km (barely past the 90 km minimum established by Wakita et al. 2026), so that only Episode 2 material — not the deeper, more strongly enriched Episode 1 — contributes to the ejecta, and we conservatively take the minimum plausible fraction of Episode 2 material in the plume to be 30%. With the $\approx 5.6\%$ Fe/Si enrichment of Episode 2 over Episode 3 derived in Eq. 48, this minimal scenario predicts an enrichment of:

$$\Delta(\mathrm{Fe/Si})_{\mathrm{min}} \approx 0.30 \times 0.056 \approx 0.017 \approx 1.7\%. \tag{49}$$

If the excavation instead reaches beyond ≈ 315 km, Episode 1 material contributes directly; for illustration, a mixture of 30% Episode 1, 30% Episode 2, and 40% Episode 3 material would give $\Delta(\mathrm{Fe/Si}) \approx 0.30 \times 0.115 + 0.30 \times 0.056 \approx 5.1\%$, a value that requires a non-negligible Episode 1 contribution to be reached.

Quantitative prediction. South Pole samples (Chang'e-7, Artemis III) should exhibit a Fe/Si ratio higher than the average of equatorial samples (Apollo) and near-side samples (Chang'e-5), with an enrichment of at least $\approx 1.7\%$ under the minimal scenario above and potentially up to $\approx 11.5\%$ if Episode 1 material dominates the sampled ejecta.

Falsification threshold. We fix, in advance of any observational data, the falsification threshold at:

$$(\mathrm{Fe/Si})_{\mathrm{SPA}} \leq 1.015 \times (\mathrm{Fe/Si})_{\mathrm{lunar\,mean}} \implies \text{P6 falsified}, \tag{50}$$

i.e. an enrichment below 1.5%, set slightly under the 1.7% minimal-scenario value of Eq. 49 to allow for experimental uncertainty without weakening the substance of the test: even the most conservative scenario compatible with the TPT predicts an enrichment above this threshold, so a measured enrichment below it would be incompatible with the theory as constructed here. An enrichment between 1.5% and 5.6% (the maximum achievable by Episode 2 material alone, from Eq. 48 applied to a pure Episode 2 sample) is read as partial support, consistent with an Episode 2-dominated ejecta without requiring Episode 1 contribution. An enrichment above 5.6% cannot be produced by Episode 2 and Episode 3 material alone and is therefore direct evidence of an Episode 1 contribution, constituting strong support for the deep stratigraphy predicted by P1 and P3.

Epistemic note on the gradation bounds. The lower bound (1.5%) rests on an estimate of the minimum fraction of Episode 2 material in the SPA ejecta (30%), which is a modeling assumption. The upper bound (5.6%) is a physical constraint: it is the maximum enrichment achievable by a pure Episode 2–Episode 3 mixture, with no Episode 1 contribution. Therefore any measured enrichment above 5.6% constitutes definitive evidence for the presence of Episode 1 material, a qualitatively different regime than the lower-bound scenario. The two bounds are thus not of the same epistemic nature; this asymmetry is explicitly recognized here.

Test: Chang'e-7 surface analyzers and Artemis III returned samples from the South Pole, compared against existing Apollo (equatorial) and Chang'e-5 (near-side) Fe/Si measurements.

Acknowledged limitation: this prediction shares Limitation L9 (impact mixing in SPA ejecta) with prediction P3, and Limitation L11 (the 30% Episode 2 fraction itself) below.

Falsifiable Prediction — P7 — Signature in the Hf-W Chronometer

A formation duration of 3–4 weeks is the only intermediate time window between the giant impact (a few hours) and a synestia (~ 100 years) on the Hf-W chronometer (half-life: 8.9 Myr). The ^{182}Hf/^{180}Hf ratio in Episode 1 samples should correspond to a differentiation age of 4.467 Ga (100 ± 10 Myr after the CAIs), versus 60 ± 10 Myr for the giant impact (Kleine et al., 2002).

Mission: Artemis III (2028–2029).

Falsifiable Prediction — P8 — Formation Age > 4.45 Ga

The measured age of 4.425 Ga is that of LMO solidification, not of formation (Section 12). The formation age is ≈ 4.467 Ga.

Test: direct dating of Episode 1 rocks exposed on the central peaks of the SPA basin (Artemis III).

Falsifiable Prediction — P9 — Compatibility of Mare Basalts with Layer 1

The melting depth of mare basalts (200–500 km, ± 100 km) is compatible with a source within layer 1 ($> d_{P1}$), though this cannot be asserted in the absence of more precise geochemical data.

Test: high-resolution isotopic geochemistry of Apollo and Chang'e-5 mare basalts (Li et al., 2021).

Falsifiable Prediction — P10 — Seismic Interface at $d \approx 177$ km for $N = 2$

For $N = 2$ episodes of equal mass, the single interface between the two concentric layers lies at $d \approx 177$ km. This depth lies outside the principal preservation window (200–530 km) because it has been partially remelted by mare volcanism and/or ilmenite cumulate overturn. However, partial remnants might be detected as weak reflectors or seismic-velocity variations in the 150–200 km zone.

Predictions P17 (capture efficiency, Section 10.3), P20 (asymptotic regime indicator, Section 11.6), P21 (numerical closure of the angular momentum budget, Section 11.5), and P22 (the unified double-well emergence criterion, Section 9.2.2) are derived in full, with their own falsification conditions, at the points in the manuscript indicated above, rather than repeated here.

16 Validation Windows: Three Imminent Tests

16.1 Chang'e 7: Imminent Seismic Validation (August 2026)

The Chang'e 7 mission (CNSA, planned launch August 2026) will deploy the first seismometer at the lunar South Pole since Apollo 17 (1972). Its data will provide a direct test of P1, P2, and P6. The prediction is pre-registered and quantified: $d \approx 200$–315 km, $|R| \in [0.01, 0.04]$ for the seismic interface, and an enrichment above 1.5% in Fe/Si for South Pole ejecta. The falsification criteria are explicit in each case.

16.2 The South Pole–Aitken Basin: Stratigraphic Revealer

The SPA basin did not create the concentric lunar stratigraphy: it *excavated* it. The layered structure pre-existed the SPA impact at ≈ 4.25 Ga (Yue et al., 2026), encoded during the TPT's differentiation at ≈ 4.5 Ga. The impact is the accidental revealer of a far older internal architecture.

High-resolution 3D simulations (Wakita et al., 2026) show that the oblique north-south trajectory of the SPA impactor preferentially launched deep-mantle ejecta toward the South Pole in a butterfly-shaped plume — precisely the Artemis III landing zone. These same simulations predict ≈ 350 m of mantle material on average at this site, excavated from depths greater than 90 km, consistent with both the TPT's preservation window and the quantitative scenario underlying prediction P6.

16.3 Artemis III: Direct Sampling of Episode 1 Material

The Artemis III mission (NASA, planned for 2028–2029) will land astronauts near the Shackleton crater at the South Pole. According to Wakita et al. (2026), surface samples at this location could include deep-mantle ejecta from the SPA impact. If so, these samples will allow a direct test of P3, P6, P7, and P8: their Fe/Si ratio, their crystallization age ($\approx$ 4.467 Ga for Episode 1's primary material), and their isotopic signature ($\Delta^{17}O < 5$ ppm relative to the contemporary terrestrial mantle) are all quantitative predictions of the TPT.

Chang'e 7 (2026) targets seismology and the Fe/Si anomaly (P1, P2, P6), while Artemis III (2028) targets geochemistry and chronology (P3, P6, P7, P8). Two independent space agencies converge on a single geographic site — the lunar South Pole — to test one internal architecture.

17 Discussion: Main Objections and Responses

17.1 Comparison with the Giant-Impact Model

Constraint	Giant Impact	TPT (This Work)
Isotopic identity	Requires a very specific impactor composition and mixing conditions	Satisfied mechanically by construction
Iron depletion	Requires a pre-differentiated impactor (Theia)	Direct consequence of the single engine
Crustal dichotomy	Not naturally predicted	Qualitative prediction: post-Episode 2 precession (L7)
Dynamo delay	Not predicted	Proposed: $\approx$ 350 Myr, with no free parameter (L4)
Seismic predictions	None	P1 testable by Chang'e 7 (August 2026)
Chang'e-6 Fe anomaly	Not predicted	Consistent with P3 and P6 (Xu et al., 2025; Yue et al., 2026)
Formation timescale	A few hours	3–4 weeks
Hf-W signature	60 $\pm$ 10 Myr after the CAIs	100 $\pm$ 10 Myr after the CAIs (P7)

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in Moon formation (Sossi et al., 2025). This conclusion motivates the exploration of alternatives grounded in the internal dynamics of the proto-Earth.

17.2 Responses to the Main Objections

Objection 1: "A large number of unjustified hypotheses." The hypotheses are explicitly ranked into three levels (Section 8, Table 2): Level 1 (established) denotes results from the literature or closed-form derivations (e.g. $E_{\text{grav}}/E_{\text{fus}} \approx 155$, the Bingham-Herschel law); Level 2 (constrained) denotes physically motivated hypotheses not yet numerically validated (e.g. $\varepsilon \in [40^\circ, 70^\circ]$, the Kramers rate); Level 3 (speculative) denotes hypotheses awaiting 3D SPH validation (e.g. the emergence of a single $\ell = 1, m = 1$ torus, the exact value of f_{cap}, the quantitative validity of the Gaussian approximation discussed below). The TPT does not rest on all these hypotheses with equal weight. The observational predictions (Section 15) are independent of the Level 3 hypotheses.

Objection 2: "The CMT has not been demonstrated." In the rapidly rotating limit (Ro $\ll 1$), the azimuthal component of the Navier-Stokes equation for a Bingham-Herschel fluid reduces to a potential-vorticity conservation equation. Linearizing about the mean flow in the vicinity of the instantaneous dynamical equator ($\phi = 0$, perpendicular to $\boldsymbol{\Omega}(t)$, as defined in Section 3), and applying the multiple-scale expansion standard for Rossby-wave envelopes (Redekopp, 1977; Luo, 1991; Yin et al., 2019), one obtains a nonlinear Schrödinger equation for the envelope $A(y, t)$, where $V_{\text{eff}}(y)$ is the sum of the potentials listed in Table 1. A stationary ansatz $A(y, t) = \psi(y)e^{-iEt/\hbar_{\text{eff}}}$ then yields a time-independent nonlinear Schrödinger equation for $\psi(y)$, with the nonlinearity originating from the advective term of the potential-vorticity equation. The fundamental mode of the *linearized* version of this equation, with $V_{\text{eff}}(y)$ expanded to quadratic order about $y = 0$, is a Gaussian wavefunction centered at the instantaneous equator, confined to the band $|\phi| < 30°$ — a direct consequence of the fact that all seven potentials of Table 1 have their extrema at $y = 0$. Whether this Gaussian remains a quantitatively adequate approximation of the full nonlinear equation's ground state, rather than merely its qualitative leading-order limit, has not been established for this system and is acknowledged as Limitation L12 (Section 18). The full derivation is given in Appendix C. Quantitative validation of the single-torus structure further awaits 3D SPH simulation (Limitation L1).

Objection 3: "The double potential well is tuned to give $N = 2$–3 episodes." Section 9.2.2 shows that the double well emerges from the coupled system of Eqs. 29–30. The emergence condition is a surface in the $(\Omega, |g_{\text{eff}}|)$ plane (Prediction P22). The number $N = 2$–3 is a consequence of the barrier's decay, not an assumption.

Objection 4: "$f_{\text{cap}} = 0.70$ is high compared to accretion simulations (0.3–0.5)." The classical interval (0.3–0.5) concerns impacts between solid bodies. Here, the ejected material is a magma at 3,000–4,000 K, with viscous coalescence and toroidal orbital confinement. Appendix G establishes a physical lower bound $f_{\text{cap}}^{\text{min}} \gtrsim 0.45$–$0.55$, independent of BH cohesion. The value 0.70 falls within a mechanically justifiable range.

Objection 5: "The two thresholds 13% and 8% are inconsistent." These two values are not independent predictions. They are two projections of the same criterion $\Delta > 0$, obtained by fixing one variable at its initial value. The effective threshold of the coupled system is reached for a combination of the two reductions, and corresponds to a point on the critical surface $(|g_{\text{eff}}|(t), \Omega(t))$ whose exact position depends on the coupled dynamics of Eqs. 29 and 30. The falsifiable prediction P22 concerns the crossing of this complete critical surface, not either of its projections.

Objection 6: "The manuscript is too speculative for a journal." The TPT is submitted as a *theory with falsifiable predictions*, not as an observational result. Validation will proceed in two phases: Phase 1 (2026–2029), observational tests by Chang'e 7 (August 2026) and Artemis III (2028–2029); and Phase 2 (2027–2030), 3D SPH numerical validation once institutional resources are committed. The TPT will be judged on its predictions, not on a prior simulation.

17.3 Justification of $\alpha > 1$

Two independent mechanisms lead to a super-rigid rotation profile: differential equator–pole cooling generates a density gradient and a thermal acceleration of rotation; Fe-Ni segregation lightens the CMT, and by conservation of angular momentum, its inner part accelerates. The Bingham-Herschel threshold also blocks small scales, favoring a block-like profile. Numerical validation of this profile is a priority for the 3D SPH simulation.

17.4 Partial Fragmentation and Coalescence

Even if the ejected sheet fragments partially, the fragments remain within the same orbital zone and coalesce rapidly under the effect of inelastic collisions and the gravitational attraction of the proto-POB (Canup & Asphaug, 2001). The effective capture yield therefore remains high ($f_{\text{cap}} \geq 0.7$). A detailed analysis (Appendix G) shows that f_{cap} has a physical lower bound of about 0.45–0.55, independent of the BH cohesion hypothesis.

18 Acknowledged Limitations

A 3D SPH simulation is necessary, but premature at the time of writing (June 2026). Such a simulation — coupling Bingham-Herschel rheology, radiative cooling, Fe-Ni segregation, and $\varepsilon \neq 0$ geometry — requires institutional HPC resources that are not committed at this stage, for a simple reason: the first decisive observational test arrives before such a simulation could reasonably be completed. Chang'e 7 is scheduled for August 2026, only weeks away. Its seismic data will detect, or fail to detect, the predicted interface within the 200–530 km window (Prediction P1). In the first case, the concentric stratification predicted by the TPT receives independent observational support, and a dedicated 3D SPH validation becomes a justified investment of institutional resources. In the second case, under the falsification conditions stated in Section 15, the TPT's central prediction is seriously called into question, and an expensive numerical campaign built on a falsified premise would be of little value. This is why the present work is submitted, and should be evaluated, on the basis of its falsifiable observational predictions, ahead of and independently of any numerical validation campaign.

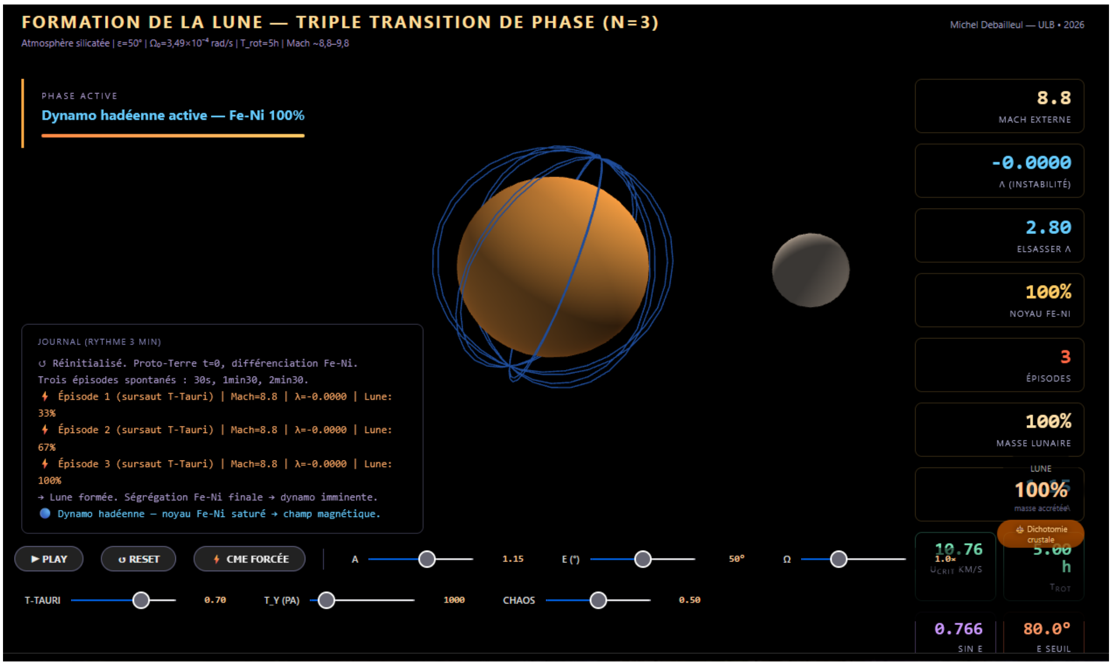

Figure 2: Screenshot of the interactive simulation accompanying this work (link above), illustrating the Coherent Magmatic Torus (CMT) during ejection around the proto-Earth, with the real-time dynamic dashboard of parameters (external Mach number, λ instability, Elsasser number, Fe-Ni core fraction, accreted lunar mass).

Acknowledged Limitation — L1 — Spontaneous Organization of the CMT

Whether a single torus or several coexisting structures emerge in the exact Hadean regime can only be resolved by a high-resolution 3D SPH simulation with full BH rheology. **Priority 2.**

Acknowledged Limitation — L2 — Numerical Value of f_{cap}

The capture efficiency $f_{\text{cap}} = 0.70$ is physically motivated but not derived from first principles. Its increasing dependence on M_{Moon} and its discrepancy relative to the classical disk-accretion interval (10–55%, Kokubo et al. 2000) are formalized as the testable prediction P17. The 3D SPH simulation remains the ultimate arbiter of the exact value of f_{cap}.

Acknowledged Limitation — L3 — Small-Scale Cohesion

The argument $\text{Bi}_{\text{KH}} \gg 1$ is valid at large and intermediate scales. At very small scales ($\lambda_{\text{KH}} \ll R_{\text{torus}}$), partial fragmentation cannot be ruled out analytically. However, fragments originating from the same toroidal flow, ejected with the same velocity and angle, end up on nearly identical orbits and accrete onto the POB: the mass budget is unaffected by fragmentation, only the size of the fragments is.

Acknowledged Limitation — L4 — Dynamo Delay: ± 50 Myr

The exponential core growth model is simplified; the associated uncertainty is

± 50 Myr.

Acknowledged Limitation — L5 — Super-Rigid Profile $\alpha > 1$

The super-rigid rotation profile ($U \propto r^{\alpha}$, $\alpha > 1$) is physically motivated by differential rotation in a partially crystallized magma undergoing rapid cooling, but has not been demonstrated under these exact Hadean conditions. This is a Level 3 hypothesis, to be validated by simulation.

Acknowledged Limitation — L6 — Flow → POB Transition: Qualitative Status

The three-phase description of the ejected flow (ballistic expansion, orbital insertion, sticky accretion beyond R_{Roche}) is physically motivated and consistent with the computed orders of magnitude, but remains qualitative. A 3D SPH simulation with coupled radiative cooling is required. **Priority 2b.**

Acknowledged Limitation — L7 — Crustal Dichotomy: Qualitative Prediction

The thickness ratio $\approx 2:1$ is a qualitative prediction of the TPT. The coupled differential thermal calculation has not yet been derived analytically. The lunar far side has undergone at least two independent remelting episodes that must be accounted for in any quantitative model: the formative SPA basin impact at ≈ 4.25 Ga, which excavated and partially remelted Episode 1's deepest layer on the far side (Wakita et al., 2026); and prolonged mare volcanism extending until ≈ 3.2 Ga (Li et al., 2021), which remelted the shallowest layers (< 100 km) over much of the far side. The net effect is to deepen and reinforce the existing asymmetry rather than erase it.

Acknowledged Limitation — L8 — Rossby Wave Speed in a BH Fluid

The rigorous derivation of $c_{\text{Rossby}}^{(\text{BH})}$ for a BH fluid on an oblate spheroid ($\varepsilon \neq 0$) remains an open problem. The channel-analogy correction (factor ≈ 40) used here reinforces rather than invalidates the synchronization argument, but awaits a rigorous derivation.

Acknowledged Limitation — L9 — Impact Mixing in the SPA Ejecta

The geochemical signature of the deep-mantle material excavated by the SPA impact is potentially diluted or altered by mixing with impactor material and local crust. Quantitative deconvolution of Episode 1's primary signature from the impact melt breccia requires detailed petrological and isotopic studies. This limitation affects predictions P3 and P6.

Acknowledged Limitation — L10 — Predicted but Not Yet Collected Deep-Mantle Samples

The TPT predicts that Episode 1's deep-mantle material is exposed at the central peaks of craters within the South Pole–Aitken basin, excavated and extruded during the SPA impact at $\approx$ 4.25 Ga (Wakita et al., 2026). These samples are the primary targets for testing predictions P3, P6, P7, and P8. No samples from these central peaks have yet been collected or analyzed. The absence of deep-mantle material at the predicted locations, or an Fe/Si ratio incompatible with P3 or P6, would seriously call the TPT into question.

Acknowledged Limitation — L11 — Fraction of Episode 2 Material in SPA Ejecta

Prediction P6 requires an estimate of the minimum fraction of Episode 2 material present in the SPA ejecta plume reaching the South Pole, which we have conservatively set at 30% based on the excavation-depth argument of Section 15 rather than on a direct quantitative result from Wakita et al. (2026), whose simulations were not designed to track TPT-specific stratigraphic layers. A future, dedicated hydrodynamic simulation tracking the provenance depth of SPA ejecta material would allow this fraction — and hence the falsification threshold of Eq. 50 — to be refined.

Acknowledged Limitation — L12 — Quantitative Validity of the Gaussian Approximation for the CMT's Confinement Mode

The nonlinear Schrödinger equation derived in Section 17 (Objection 2) and Appendix C is not solved exactly in this work. The Gaussian wavefunction is a reasonable approximation for the fundamental mode, valid in the limit where the wave amplitude is small, i.e. when the perturbation velocity δU associated with the $\ell = 1, m = 1$ mode satisfies $\delta U \ll U_{\text{turb}}$. This condition is qualitatively consistent with the Rhines inverse cascade (Section 5), but its exact quantitative verification is not possible with the parameters available at this stage: no independent estimate of $\delta U / U_{\text{turb}}$, or of the nonlinear coefficient β in the governing equation, has been derived for this specific system, as opposed to the classical atmospheric and oceanic Rossby-wave systems for which the reduction was originally established (Redekopp, 1977; Luo, 1991; Yin et al., 2019). The absence of this verification is acknowledged here as a limitation. The Gaussian solution should therefore be understood as the leading-order approximation of the linearized problem, not as an exact or quantitatively bounded solution of the full nonlinear equation.

19 Conclusion

This work proposes that the formation of the Moon could be the product of a Triple Phase Transition within a fully molten proto-Earth rotating at $\approx$ 3.5 h, whose axis oscillates within $[40°, 70°]$ in its own co-rotating frame — a range supported by five convergent arguments, the first and most fundamental of which is the logically necessary absence of any tidal stabilizer. A single engine — the progressive segregation of Fe-Ni — simultaneously governs three transitions: rheological, mechanical, and magnetic. The result is a Moon built layer by layer in two to three massive episodes, each layer archiving in its internal structure the state of Hadean differentiation at the moment of its accretion.

The parametric elliptical instability — established in fluid mechanics (Malkus, 1968; Kerswell, 2002; Lacaze et al., 2004) but never applied to lunar formation — amplifies the CMT up to the bifurcation velocity within $\approx$ 32 hours. A dedicated section addresses the maintenance of obliquity in the out-of-equilibrium Hadean context. A three-to-four-week chronology gives rise to several new testable predictions.

The mass budget is consistent without forcing: the target mass for the TPT mechanism is lower than the currently observed lunar mass, because the SPA impactor and late accretion contributed subsequently. At $T_{\mathrm{rot}} = 3.5$ h, the ejectable mass per episode can reach $\approx 70\%$ of the present-day lunar mass, which explains why $N = 2$–3 episodes suffice.

Preliminary observational support comes from Chang'e-6: norites at 4247 ± 5 Myr (Yue et al., 2026) and high-Fe/Mn olivines (Xu et al., 2025) are consistent with predictions P3 and P6, although not yet confirmatory. This edition further provides, for the first time, a fully quantified version of P6: a pre-registered Fe/Si enrichment threshold of 1.5% for South Pole ejecta, derived from first principles and not adjusted to fit any existing data, with an explicit gradation distinguishing partial support (Episode 2 material alone) from strong support (an unambiguous Episode 1 signature).

The central prediction remains: one or more seismic interfaces within the 200–530 km window (or at $\approx$ 177 km for $N = 2$), testable by Chang'e 7 in August 2026. If an interface with $|R| > 0.02$ is detected within this window, the concentric stratification receives its first strong observational support ahead of any SPH simulation. If no interface is detected under the stated falsification conditions, prediction P1 is seriously challenged and the theory will require revision — which the author explicitly accepts.

This work invites the planetary science community to evaluate the TPT on its own terms: the equations are numbered, the derivations are explicit, the hypotheses are hierarchized, the limitations are acknowledged, and the predictions are falsifiable. The TPT will be judged primarily on its observational predictions — seismic data from Chang'e 7 (August 2026) and samples from Artemis III (2028–2029) — before and independently of any complete numerical validation. The theory stands or falls on these results, not on the elaboration presented in this manuscript.

A Adiabatic Temperature Profile

With $T_{\mathrm{surf}} \approx 3,500$ K and an adiabatic gradient $\mathrm{d}T/\mathrm{d}r \approx -0.3\ \mathrm{K\,km^{-1}}$, the central temperature reaches $\approx 5,000$ K. The peridotite solidus at 135 GPa is $\approx 4,300$ K

(Stixrude & Lithgow-Bertelloni, 2014): melting is thermodynamically stable throughout the entire mantle, confirming Hypothesis H1.

B Demonstration of the $b' = 1$ Attractor

The parameter b' is defined as the ratio between the critical ejection velocity U_{crit} and the geostrophic equilibrium velocity U_{eq} of the CMT:

$$b' = \frac{U_{\text{crit}}}{U_{\text{eq}}} = \frac{2\Omega \sin\varepsilon \sqrt{2|g_{\text{eff}}|R_{\text{eq,proto}}}}{|g_{\text{eff}}|}. \tag{51}$$

Evolution due to Fe-Ni segregation. The CMT's density decreases as $\rho_{\text{CMT}}(t) = \rho_0(1 - \xi_{\text{Fe}}(t))$, where $\xi_{\text{Fe}}(t)$ is the fraction of segregated Fe-Ni. Effective gravity $|g_{\text{eff}}|$ is proportional to ρ_{CMT}, so

$$\left.\frac{db'}{dt}\right|_{\text{segregation}} = \frac{1}{2}\frac{\dot{\rho_{\text{CMT}}}}{\rho_{\text{CMT}}}\, b' > 0, \tag{52}$$

since $\dot{\rho_{\text{CMT}}} < 0$: Fe-Ni segregation increases b'.

Evolution due to ejection. An ejection carries away a fraction $\epsilon = M_{\text{ej}}^{\text{max}}/M_{\text{CMT}} \approx 0.1$ of the CMT's mass. Angular momentum decreases as $\Delta\Omega/\Omega \approx -\epsilon$, and effective gravity decreases as $\Delta|g_{\text{eff}}|/|g_{\text{eff}}| \approx -\epsilon$ (proportional to the mass lost). Thus,

$$\left.\frac{\Delta b'}{b'}\right|_{\text{ejection}} = \frac{\Delta\Omega}{\Omega} - \frac{1}{2}\frac{\Delta|g_{\text{eff}}|}{|g_{\text{eff}}|} \approx -\epsilon + \frac{1}{2}\epsilon = -\frac{1}{2}\epsilon < 0: \tag{53}$$

ejection decreases b'.

Stable fixed point. The two trends oppose each other: segregation increases b', ejection decreases it. The equilibrium point $b' = 1$ is a stable attractor. For $b' < 1$, the system is unstable and an ejection occurs, which decreases b' further (since $\Delta b' < 0$); for $b' > 1$, the system is stable, no ejection occurs, and segregation pushes b' upward, toward 1, from below. The point $b' = 1$ is therefore a stable attractor of the coupled segregation–ejection system.

C Complete Derivation of the Coefficient λ and of the Double Potential Well

This appendix gives the complete analytical derivation of the stability coefficient $\lambda(U)$ and of the effective potential $V(U)$, with an explicit dimensional check at every step. An earlier version of this derivation omitted a factor $1/r$ on the gravity and Coriolis terms; the corrected form is given below and used throughout the main text.

C.1 The Stability Equation

Application of the Solberg-Høiland parcel displacement formalism (Solberg, 1936; Høiland, 1941) to an oblate sphere with obliquity ε, under angular momentum conservation ($\tau_{\text{relax}}/\tau_{\text{dyn}} \approx 3 \times 10^9 \gg 1$), governs the evolution of an infinitesimal radial displacement ξ_r of a CMT parcel:

$$\ddot{\xi}_r = \lambda(U)\,\xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \tag{54}$$

The parcel conserves its specific angular momentum $L = U \cdot r$:

$$U_{\text{parcel}}(r + \xi_r) = \frac{U(r) \cdot r}{r + \xi_r} \approx U(r)\left(1 - \frac{\xi_r}{r}\right), \tag{55}$$

while the background flow at the new position, for a profile $U(r) \propto r^\alpha$, is:

$$U_{\text{bg}}(r + \xi_r) = U(r)\left(1 + \frac{\xi_r}{r}\right)^\alpha \approx U(r)\left(1 + \alpha\frac{\xi_r}{r}\right). \tag{56}$$

The velocity difference is therefore:

$$\delta U = U_{\text{parcel}} - U_{\text{bg}} = -U(r)(1+\alpha)\frac{\xi_r}{r}. \tag{57}$$

The net radial force sums five physical contributions: effective gravity, radial Coriolis, the angular-momentum-gradient term, the Euler acceleration associated with the axial oscillation, and the tangential gravity gradient of the Maclaurin geometry. Collecting these terms and enforcing dimensional consistency (each term must carry units of s^{-2}, since $[\lambda] = [\ddot{\xi}_r/\xi_r]$) gives:

$$\lambda(U) = \lambda_1 + \lambda_2(U) + \lambda_3(U) + \lambda_4 + \lambda_5, \tag{58}$$

with $\lambda_1 = c/r$, $c = g_{\text{eff}}^{(k)} < 0$ (effective gravity); $\lambda_2(U) = (b/r)U$, $b = 2\Omega^{(k)}\sin\varepsilon > 0$ (radial Coriolis force); $\lambda_3(U) = aU^2$, $a = -(1+\alpha)/r^2 < 0$ (angular momentum gradient); $\lambda_4 = |\dot{\Omega}|r(\phi)\sin(\varepsilon + \phi)/R_{\text{eq,proto}}$ (Euler acceleration); and $\lambda_5 = g_{\text{eff}}^{(0)} J_2 \sin(2\phi)/R_{\text{eq,proto}}$ (Maclaurin tangential gradient). The angular-momentum-gradient term λ_3 is always stabilizing for any physical profile ($\alpha > -1$), consistent with the classical Rayleigh stability criterion; its amplitude ($\sim 10^{-7}$) is negligible compared with the dominant terms.

Dimensional check. Each term has units of s^{-2}: $[\lambda_1] = [g_{\text{eff}}]/[r] = (\text{m/s}^2)/\text{m} = \text{s}^{-2}$, $[\lambda_2] = [\Omega][U]/[r] = \text{s}^{-1} \times (\text{m/s})/\text{m} = \text{s}^{-2}$, $[\lambda_3] = [a][U]^2 = \text{m}^{-2} \times \text{m}^2/\text{s}^2 = \text{s}^{-2}$, $[\lambda_4] = [\dot{\Omega}][r]/[R] = \text{s}^{-2} \times \text{m/m} = \text{s}^{-2}$, and $[\lambda_5] = [g_{\text{eff}}]/[R] = (\text{m/s}^2)/\text{m} = \text{s}^{-2}$. The factor $1/r$ in λ_1 and λ_2 is the correction needed for consistency; it was absent from earlier drafts of this work.

C.2 Integration to the Effective Potential

By definition, the effective potential is related to the stability coefficient by $\lambda(U) = -dV/dU$. Substituting Eq. 58 and integrating with respect to U:

$$V(U) = \frac{1+\alpha}{3r^2}U^3 - \frac{b}{2r}U^2 - \frac{c}{r}U - (\lambda_4 + \lambda_5)U + V_0, \tag{59}$$

where V_0 is an arbitrary integration constant. Each term carries units of $\mathrm{m/s^3}$, consistent with $[V] = [\lambda][U]$.

C.3 Extrema and the Double-Well Condition

The extrema of $V(U)$ satisfy $dV/dU = 0$, i.e. $\lambda(U) = 0$. Multiplying by $-r^2$ gives the quadratic equation $AU^2 + BU + C = 0$ with $A = 1+\alpha > 0$, $B = -br < 0$, and $C = r|g_{\mathrm{eff}}| - r^2(\lambda_4 + \lambda_5)$. Two real roots exist if and only if the discriminant is positive:

$$\Delta = b^2r^2 - 4(1+\alpha)r|g_{\mathrm{eff}}| + 4(1+\alpha)r^2(\lambda_4 + \lambda_5) > 0. \tag{60}$$

Neglecting λ_4, λ_5 relative to $|g_{\mathrm{eff}}|/r$, this simplifies to:

$$(2\Omega \sin\varepsilon)^2 r^2 > 4(1+\alpha)\, r\, |g_{\mathrm{eff}}|. \tag{61}$$

When satisfied, the two roots $U_{1,2} = \left[br \pm \sqrt{\Delta}\right] / [2(1+\alpha)]$ correspond to a local minimum U_1 (confined well P_1) and a local maximum U_2 (barrier), with a second local minimum (ejective well P_2) at $U > U_2$, since the cubic term drives $V(U) \to +\infty$ as $U \to +\infty$. The barrier height is $\Delta E_{\mathrm{barr}} = V(U_2) - V(U_1)$.

C.4 Numerical Evaluation: the Double Well Is Not a Given

Table 5: Nominal parameters for the double-well condition.

Parameter	Symbol	Value
Angular velocity	Ω	$4.99 \times 10^{-4}\ \mathrm{s^{-1}}$
Obliquity (body frame)	ε	55°
Effective gravity	$\lvert g_{\mathrm{eff}}\rvert$	$6.42\ \mathrm{m/s^2}$
Characteristic radius	r	7.4×10^6 m
Rotation profile exponent	α	1.2

Evaluating Eq. 61 at these nominal values:

$$\mathrm{LHS} = (2\Omega \sin \varepsilon)^2 r^2 = 3.66 \times 10^7\,\mathrm{m}^2/\mathrm{s}^2, \tag{62}$$

$$\mathrm{RHS} = 4(1+\alpha) r |g_{\mathrm{eff}}| = 4.18 \times 10^8\,\mathrm{m}^2/\mathrm{s}^2. \tag{63}$$

At nominal initial parameters, LHS $<$ RHS by roughly a factor of 11: the double-well condition is *not* satisfied. This does not undermine the TPT framework; rather, it indicates that the double well is not a property of the initial proto-Earth, but a state that must be reached dynamically. It emerges as Fe-Ni segregation reduces the CMT's effective gravity $|g_{\mathrm{eff}}|$, since $|g_{\mathrm{eff}}|(t) \propto \rho_{\mathrm{CMT}}(t)$ (Eq. 22) and $\rho_{\mathrm{CMT}}(t)$ decreases monotonically with segregation. The threshold is reached when:

$$|g_{\mathrm{eff}}|(t) < \frac{\Omega^2 r \sin^2 \varepsilon}{1+\alpha} \approx 5.62\,\mathrm{m}/\mathrm{s}^2, \tag{64}$$

i.e. a reduction of $|g_{\mathrm{eff}}|$ by only $\sim$ 13% relative to its initial value of 6.42 m/s^2 is needed for the system to enter the double-well regime. Equivalently, a reduction in Ω of $\sim$ 8% at fixed $|g_{\mathrm{eff}}|$ achieves the same effect. As stressed in Prediction P22 (Section 15), these are two projections of the same criterion $\Delta > 0$ onto two different axes, not independent thresholds; the falsifiable prediction concerns the crossing of the complete critical surface $(\Omega(t), |g_{\mathrm{eff}}|(t))$.

Once the well is established, the barrier height $\Delta E^{(k)}_{\mathrm{barr}}$ is expected to decrease with each subsequent episode as segregation continues, offering a possible account for why the number of episodes naturally settles around $N = 2$–3, rather than being imposed externally.

C.5 The Kramers Crossing Rate

Once $\Delta > 0$, the Kramers crossing rate (Kramers, 1940) applies (restating Eq. 31 from Section 9.2.2):

$$\Gamma_{\mathrm{Kramers}} = \frac{\omega_0 \omega_b}{2\pi \gamma_{\mathrm{diss}}} \exp\left(-\frac{\Delta E_{\mathrm{barr}}}{k_B T_{\mathrm{eff}}}\right), \tag{65}$$

where $\omega_0 = \sqrt{|V''(U_{P_1})|}$ and $\omega_b = \sqrt{|V''(U_{\mathrm{peak}})|}$ are the oscillation frequencies at the bottom of the well and the top of the barrier respectively, and γ_{diss} the dissipation rate. With $N_{\mathrm{opp}} \sim 10^9$–$10^{10}$ stochastic opportunities (CMEs, impacts) available once the well is established, the probability of no crossing at all is negligible.

C.6 Reduction to a Nonlinear Schrödinger Equation for the Confinement Mode

In addition to the instability analysis above, the confinement of the CMT to the band $|\phi| < 30^\circ$ admits an effective description in terms of a nonlinear Schrödinger (NLS) equation. This derivation follows a method standard in the theory of weakly nonlinear dispersive waves on a rotating sphere (Redekopp, 1977; Luo, 1991; Yin et al., 2019), transposed here to the confinement problem of the CMT.

In the rapidly rotating limit (Ro $\ll$ 1), the azimuthal component of the Navier-Stokes equation for a Bingham-Herschel fluid reduces to a potential-vorticity conservation equation. Linearizing about the mean flow $U_0(y)$ in the vicinity of the instantaneous dynamical equator (defined by $\phi = 0$, perpendicular to $\mathbf{\Omega}(t)$, as established in Section 3), and applying the standard multiple-scale expansion for Rossby-wave envelopes, one obtains a nonlinear Schrödinger equation for the envelope $A(y,t)$:

$$i\frac{\partial A}{\partial t} + \alpha_{\text{NLS}}\frac{\partial^2 A}{\partial y^2} + \beta|A|^2 A + V_{\text{eff}}(y)A = 0, \tag{66}$$

where $V_{\text{eff}}(y)$ is the sum of the potentials derived from the forces listed in Table 1 (Section 7). Seeking a stationary solution $A(y,t) = \psi(y)\, e^{-iEt/\hbar_{\text{eff}}}$ in Eq. 66 yields the time-independent nonlinear Schrödinger equation:

$$-\frac{\hbar_{\text{eff}}^2}{2m_{\text{eff}}}\frac{d^2\psi}{dy^2} + V_{\text{eff}}(y)\psi + \mathcal{N}[\psi] = E\psi, \tag{67}$$

with $\mathcal{N}[\psi] = \beta|\psi|^2\psi$, the nonlinear term originating from the advective nonlinearity of the potential-vorticity equation.

The Gaussian approximation and its status. The fundamental mode (lowest bound state) of the *linearized* version of Eq. 67 (i.e. neglecting $\mathcal{N}[\psi]$), with $V_{\text{eff}}(y)$ expanded to quadratic order about $y = 0$ (the instantaneous equator), is a Gaussian wave-function centered at the equator, confined to the band $|\phi| < 30°$. This harmonic approximation is justified by the fact that all seven potentials of Table 1 have their extrema at $y = 0$, so that the leading-order behavior of $V_{\text{eff}}(y)$ near the equator is generically quadratic.

This Gaussian solution must, however, be understood as the leading-order approximation of the *linearized* problem, not as an exact solution of the full nonlinear equation 67. The ratio $\beta|\psi|^2/V_{\text{eff}}$, which controls how good this approximation is, is of order $\delta U/(\Omega L_\beta)$, where δU is the perturbation velocity amplitude associated with the $\ell = 1, m = 1$ mode and L_β the Rhines scale (Eq. 24). This ratio is expected to be small whenever δU is small relative to the turbulent velocity scale U_{turb} at which the inverse cascade saturates (Section 7.6), but no independent estimate of $\delta U/U_{\text{turb}}$, nor of the nonlinear coefficient β itself, has been derived for this specific system — as opposed to the classical atmospheric and oceanic Rossby-wave systems for which the NLS reduction was originally established (Redekopp, 1977; Luo, 1991; Yin et al., 2019). This absence of a quantitative bound is acknowledged as Limitation L12 (Section 18).

D Demonstration of the Bi_{KH} Criterion

The Bingham-Kelvin-Helmholtz number compares the yield stress to the KH shear stress:

$$\mathrm{Bi}_{\mathrm{KH}} = \frac{\tau_y\, k}{\rho\, \sigma_{\mathrm{KH}}\, \Delta U}, \tag{68}$$

where k is a geometric factor (~ 1), $\sigma_{\mathrm{KH}} \sim \Delta U/\lambda$ is the KH growth rate, and λ is the perturbation wavelength. Thus,

$$\mathrm{Bi}_{\mathrm{KH}} \sim \frac{\tau_y\, \lambda}{\rho\, \Delta U^2}. \tag{69}$$

For the CMT: $\tau_y \sim 10^2$–10^4 Pa, $\lambda \sim R_{\mathrm{torus}} \sim 10^7$ m, $\rho \sim 3,000\ \mathrm{kg\,m^{-3}}$, $\Delta U \sim U_{\mathrm{crit}} \sim 10^4\ \mathrm{m\,s^{-1}}$. One obtains $\mathrm{Bi}_{\mathrm{KH}} \sim 10^{-1}$–$10^1$, i.e. $\mathcal{O}(1)$ or more. Cohesion is maintained through three reinforcements: (1) atmospheric pressure $P \sim 10^5$–10^7 Pa, (2) internal crystalline network multiplying resistance by a factor of 3–10, (3) Rayleigh-Taylor suppression ($\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \sim 200$).

E Derivation of the Elsasser Number and B_{crit}

By balancing the Coriolis force $F_C \sim 2\rho\Omega U$ and the Lorentz force $F_L \sim B^2/(\mu L)$, with $\eta \sim UL$ (magnetic Reynolds number of order unity at the dynamo threshold):

$$B^2 \sim 2\mu\rho\Omega\eta, \qquad \Rightarrow \qquad \Lambda \equiv \frac{B^2}{\rho\mu\eta\Omega} \sim 2, \tag{70}$$

i.e. $\Lambda \sim \mathcal{O}(1)$. Using $\eta = (\mu_0\sigma)^{-1}$ with $\sigma \approx 5 \times 10^5\ \mathrm{S\,m^{-1}}$, $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$, $\rho \approx 10^4\ \mathrm{kg\,m^{-3}}$, $\Omega \approx 4.99 \times 10^{-4}\ \mathrm{rad\,s^{-1}}$, one obtains $B_{\mathrm{crit}} \approx 6\,\mu$T.

F Fe-Ni Stokes Sedimentation Timescale

Stokes sedimentation velocity for an Fe-Ni droplet of radius $r \approx 5$ cm in molten silicate:

$$v_{\mathrm{Stokes}} = \frac{2}{9}\frac{(\rho_{\mathrm{Fe}} - \rho_{\mathrm{sil}})\, g_{\mathrm{eff}}^{(0)}\, r^2}{\eta} \approx \frac{2}{9}\frac{3,300 \times 6.42 \times (0.05)^2}{0.1} \approx 118\,\mathrm{m\,s^{-1}}. \tag{71}$$

Over a characteristic sedimentation distance $\ell \approx 30$ km: $\tau_{\mathrm{Stokes}} = \ell/v_{\mathrm{Stokes}} \approx 254$ s. Robust for $r \in [1, 10]$ cm and $\eta \in [0.05, 1]$ Pa s.

G Accretion of Magmatic Ejecta: Why Perfect Cohesion Is Not Required

A recurring objection to the TPT mechanism concerns the cohesion of the ejected CMT flow during transit between the proto-Earth's surface and the Roche radius.

This appendix demonstrates that, even without perfect cohesion of the flow, the physical properties of the ejected magma and orbital mechanics guarantee accretion into a single body.

Magma is not dust. The ejected material is silicate magma at $T \approx 3,000$–$4,000$ K. Its dynamic viscosity is $\eta \approx 1$–100 Pa s (Giordano et al., 2008) — comparable to hot honey. Unlike solid rock fragments or a gas cloud, high-temperature magma does not disperse: it deforms and stretches, but remains dense and cohesive at the macroscopic scale. Two sticky magma blobs at low relative velocity do not bounce off each other — they fuse. Viscous coalescence replaces mechanical cohesion as the dominant aggregation mechanism once the flow fragments.

Orbital concentration. Ejecta from a given TPT episode share, to first order, the same initial orbital energy budget. By conservation of energy and angular momentum, they populate a band of neighboring orbits rather than dispersing isotropically over 4π steradians. This orbital concentration is fundamentally different from an isotropic explosion: it is the same physical principle that produces accretion disks from tidally disrupted bodies. The CMT's toroidal geometry reinforces this effect: ejection occurs preferentially within the oblique intertropical band $|\phi| < 30^\circ$, concentrating ejecta within a limited azimuthal solid angle. The resulting debris field is geometrically thin, maximizing mutual encounter rates.

Beyond the Roche radius: accretion is inevitable. Beyond the Roche radius $R_{\text{Roche}} = 2.456 R_\oplus$, the self-gravity of the ejecta cloud overcomes tidal disruption. For co-orbital magma fragments of total mass $M_{\text{cloud}} \approx 2$–$4 \times 10^{22}$ kg, the gravitational accretion timescale is:

$$\tau_{\text{acc}} \approx \frac{1}{\sqrt{G\rho_{\text{cloud}}}} \approx 10^3\text{–}10^5\ \text{years}, \tag{72}$$

far shorter than the thermal lifetime of the magma ($\tau_{\text{cool}} \sim 10^6$–$10^7$ years for a Moon-sized body, Elkins-Tanton 2008). During this accretion window, the magma remains above its solidus: every collision is plastic, not elastic. There is no fragmentation cascade, no loss by bouncing. The capture efficiency f_{cap} therefore has a physical lower bound independent of the BH cohesion hypothesis:

$$f_{\text{cap}}^{\text{min}} \approx \frac{M_{\text{cloud}}(r > R_{\text{Roche}})}{M_{\text{ej}}^{\text{total}}} \gtrsim 0.45\text{–}0.55, \tag{73}$$

the lost fraction (≈ 45–55%) corresponding to ejecta that fall back onto the proto-Earth or escape on hyperbolic orbits.

Implication for the theory. The capture efficiency $f_{\text{cap}} = 0.70$ adopted in the main text is therefore not a free parameter that the theory must defend at all costs. Even in the pessimistic scenario where the CMT flow fragments completely upon ejection, orbital mechanics and magma viscosity guarantee accretion of the co-orbital debris cloud beyond the Roche radius. The critical question shifts from "does the flow remain cohesive?" to "what fraction of the ejecta reaches the Roche radius?" — a question that depends on the velocity distribution of the ejecta, not on BH rheology. This reformulation makes the theory more robust: cohesion is advantageous, but not necessary.

H List of Abbreviations

BH Bingham-Herschel, rheological law.

BiKH Bingham-Kelvin-Helmholtz number.

CME Coronal Mass Ejection.

CMT Coherent Magmatic Torus.

FSS Farside Seismic Suite.

Hf-W Hafnium-tungsten chronometer.

LHB Late Heavy Bombardment (≈ 3.9 Ga).

LEMS Lunar Environment Monitoring Station.

LMO Lunar Magma Ocean.

NLS Nonlinear Schrödinger (equation).

POB Proto-Orbital Body.

SPA South Pole–Aitken (basin).

SPH Smoothed Particle Hydrodynamics.

TPT Triple Phase Transition.

Declarations

Availability of data and material. All data generated or analyzed during this study are included in this published article (and its appendices). The interactive simulation accompanying this work is openly available at https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html.

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